BIXOLON

API Reference Guide

Label Printer SDK

iOS

Ver. 3.16

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Caution

Some semiconductor devices are easily damaged by static electricity. You should turn the printer “OFF”, before you connect or remove the cables on the rear side, in order to guard the printer against the static electricity. If the printer is damaged by the static electricity, you should turn the printer “OFF”.

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1. About This Manual

This SDK Manual contains the description of the library API that is required for the development of iOS applications.

1-1 Supported iOS Version

• iOS 12.0 and later.

1-2 List of Supported Printer / Interface

Model	Wi-Fi	Ethernet	Bluetooth	BLE
SLP-DX420	O	O	O	O (BT-MSOIII)
SLP-DX423	O	O	O	O (BT-MSOIII)
SLP-DX220	O	O	O	O (BT-MSOIII)
SLP-DX223	O	O	O	O (BT-MSOIII)
SLP-TX420	O	O	O	O (BT-MSOIII)
SLP-TX423	O	O	O	O (BT-MSOIII)
SLP-TX220	O	O	O	O (BT-MSOIII)
SLP-TX223	O	O	O	O (BT-MSOIII)
SLP-DL410	O	O	O	O (BT-MSOIII)
SLP-DL413	O	O	O	O (BT-MSOIII)
SLP-TX400	O	O	X	O (BT-MSOIII)
SLP-TX403	O	O	X	O (BT-MSOIII)
SRP-770III	O	O	X	X
SRP-E770III	X	O	X	X
XT5-40(RFID)	O	O	O	O
XT5-43(RFID)	O	O	O	O
XT5-46(RFID)	O	O	O	O
XT6-60N	O	O	O	O
XT6-63N	O	O	O	O
XL5-40CT	O	O	O	O
XL5-43CT	O	O	O	O
XD5-40d	O	O	O	O
XD5-40IId	O	O	O	O
XD5-43d	O	O	O	O
XD5-43IId	O	O	O	O
XD5-40t(RFID)	O	O	O	O
XD5-40IIt(RFID)	O	O	O	O
XD5-43t(RFID)	O	O	O	O
XD5-43IIt(RFID)	O	O	O	O
SPP-L3000	O	X	O	O
SPP-L310	O	X	O	O

SPP-L410	O	X	O	O
XM5-30(RFID)	O	X	O	O
XM7-40(RFID)	O	O (Cradle required (optional))	O	O
XM7-20	O	O (Cradle required (optional))	O	O
XM7-30	O	O (Cradle required (optional))	O	O
SRP-S3000	O	O	X	X
XT3-40	O	O	O	O
XT3-43	O	O	O	O
BT3-40	O	O	O	O
BT3-43	O	O	O	O
BT5-40(RFID)	O	O	O	O
BT5-43(RFID)	O	O	O	O
BT5-46(RFID)	O	O	O	O
BD5-40d	O	O	O	O
BD5-43d	O	O	O	O
BD5-40t(RFID)	O	O	O	O
BD5-43t(RFID)	O	O	O	O
XC3-40d	X	X	O	O
XC5-40d	O	O	O	O
XC5-43d	O	O	O	O

※BLE: Bluetooth Low Energy

1-3 Available range of X/Y coordinates for each model

Model	Min Width	Max Width
SLP-DX420	0	864
SLP-DX423	0	1248
SLP-DX220	0	432
SLP-DX223	0	672
SLP-TX420	0	864
SLP-TX423	0	1248
SLP-TX220	0	432
SLP-TX223	0	672
SLP-DL410	0	864
SLP-DL413	0	1248
SLP-TX400	0	864
SLP-TX403	0	1248
SRP-770III	0	832
SRP-E770III	0	832
XT5-40(RFID)	0	864
XT5-43(RFID)	0	1248
XT5-46(RFID)	0	2496
XT6-60N	0	1344
XT6-63N	0	1984
XL5-40CT	0	864
XL5-43CT	0	1248
XD5-40d	0	864
XD5-40IId	0	864
XD5-43d	0	1248
XD5-43IId	0	1248
XD5-40t(RFID)	0	864
XD5-40IIt(RFID)	0	864
XD5-43t(RFID)	0	1248
XD5-43IIt(RFID)	0	1248
SPP-L3000	0	576
SPP-L310	0	576
SPP-L410	0	832
XM5-30(RFID)	0	576
XM7-40(RFID)	0	832
XM7-20	0	384
XM7-30	0	576
SRP-S3000	0	576
XT3-40	0	864
XT3-43	0	1248
BT3-40	0	864

BT3-43	0	1248
BT5-40(RFID)	0	832
BT5-43(RFID)	0	1248
BT5-46(RFID)	0	2496
BD5-40d	0	864
BD5-43d	0	1248
BD5-40t(RFID)	0	864
BD5-43t(RFID)	0	1248
XC3-40d	0	864
XC5-40d	0	864
XC5-43d	0	1248

※ The maximum Y Coordinate is dependent on paper length

1-4 List of Supported Methods

Method		Remarks
General	*open*	-
	close	-
Search	lookupPrinterWithCount: interval:	-
	CancelToLookupPrinter	-
	getPairedDevices	-
	getSerchedDevices	-
Connection	*connectWithAddress:port:*	-
	connectWithTarget:timeout:	-
	connectWithSerialNumber:	-
	isConnected	-
	disconnect	-
	disconnectWithTimeout:	-
Print	*doPrint*	-
Text	*setCurrentEncoding*	-
	drawTextDeviceFont: xPosition: yPosition: fontSelection: fontWidth: fontHeight: rightSideCharacterSpacing: fontRotation: reverse: bold: textAlignment:	-
	drawTextVectorFont: xPosition: yPosition: fontSelection: fontWidth: fontHeight: rightSideCharacterSpacing: fontRotation: reverse: bold: italic: textAlignment:	-

Barcode	*drawBarcode1D:* xPosition: yPosition: barcodeType: widthNarrow: widthWide: height: hri: quietZoneWidth: rotation:	-
	drawBarcodeMaxiCode: xPosition: yPosition: mode:	-
	drawBarcodePDF417: xPosition: yPosition: maximumRowCount: maximumColumnCount: errorCorrectionLevel: dataCompressionMethod: printBarcodeText: barcodeOriginPoint: moduleWidth: barHeight: rotation:	-
	drawBarcodeQRCode: xPosition: yPosition: barcodeSize: model: errorColectionLevel: rotation:	-
	drawBarcodeDataMatrix: xPosition: yPosition: barcodeSize: reverse: rotation:	-
	drawBarcodeAztec: xPosition: yPosition: barcodeSize: extendedChannel:	-

	errorCorrectionLevel: menuSymbol: numberOfSymbols: optionalID: rotation:	
Barcode	drawBarcodeCode49: xPosition: yPosition: widthNarrow: widthWide: height: hri: startingMode: rotation:	-
	drawBarcodeCodaBlock: xPosition: yPosition: widthNarrow: widthWide: height: securityLevel: numberOfCharactersPerrow: mode: numberOfRowToEncode:	-
	drawBarcodeMicroPDF: xPosition: yPosition: moduleWidth: barcodeHeight: mode: rotation:	-
	drawBarcodeIMB: xPosition: yPosition: printBarcodeText: rotation:	-
	drawBarcodeMSI: xPosition: yPosition: widthNarrow: widthWide: height: checkDigitSelection: printCheckDigitInHRI: hri: rotation:	-

	drawBarcodePlessey: xPosition: yPosition: widthNarrow: widthWide: height: printCheckDigit: hri: rotation:	-
Barcode	drawBarcodeTLC39: xPosition: yPosition: widthNarrow: widthWide: height: rowHeightOfMicroPDF417: narrowWidthOfMicroPDF417: rotation:	-
	drawBarcodeRSS: xPosition: yPosition: barcodeType: magnification: separatorHeight: barcodeHeight: segmentWidth: rotation:	-
	drawBarcodeGS1DataBar: xPosition: yPosition: barcodeType: magnification: separatorHeight: barcodeHeight: segmentWidth: rotation:	-
Block & Circle	drawBlock: startPosY: endPosX: endPosY: option: thickness:	-
	drawCircle: startPosY:	-

	sizeSelection: multiplier:	
Image	drawImage: startPosX: startPosY: width: useDithering:	-
Status	checkPrinterStatus:	-
Cut	setAutoCutter: cuttingPeriod:	Cutter mounted model only
Information	getSDKVersionString	-
	getReleasedDateString	-
	getDeviceFontList	-
	getVectorFontList	-
	getDrawBlockOptions	-
	getDrawCircleSizes	-
	getBarcodeType1D	-
	getBarcodeHRI	-
	getDpi	-
	getMaxWidth	-
	getModelName	-
	getFirmareVersion	-
Printer Setting	setOrientation	-
	setICS: codepage	-
	setPrintingType	-
	setMargin: verticalMargin	-
	setLength setPrintingType	-
	setRewinder	Rewinder mounted model only
	setSpeed	-
	setOffset	-
	setDensity	-
	setCuttingPosition	-
	setWidth	-
Print RawData	printRawData	-

2. Label Printer SDK Class Reference

Inherits from	NSObject
Framework	libLabelPrinterSDK.a
Declared	LabelPrinterSDK.h

2-1 Over view

• libLabelPrinterSDK Class is the main object that controls the printer operation.

2-2 Methods

2-2-1 Open

This method initializes the settings to use the LabelPrinterSDK class.

[Declaration]

-(long) open;

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_OPEN_ALREADY	0XF003	SDK is already open



This method should be called before calling other methods in the Label Printer SDK.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

__SDK_RESULT_CODES_ result = [sdk open];

If (result == _SDK_RESULT_SUCCESS) {
        …
} else {
        …
}
```

2-2-2 Close

This method terminates the use of the LabelPrinterSDK class.

[Declaration]
-(long) close;

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_NO_OPEN	0XF002	SDK is not open



All resources inside the LabelPrinterSDK will be released when this method is called.
This method should be called to end the use of the printer.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

…

__SDK_RESULT_CODES_ result = [sdk close];

…

If (result == _SDK_RESULT_SUCCESS) {
      …
} else {
      …
}
```

2-2-3 Lookup Printer With Count

It searches for connectable printers.

[Declaration]
-(void) lookupPrinterWithcount:(CGFloat) lookupCount
Interval:(CGFloat) interval

[Parameters]
lookupCount: It specifies the number of printer search attempts. (Range: 1~255)
interval: It specifies the interval between printer search attempts.
(Range: 0.2~10(seconds))

If the parameter is (lookupCount=10, interval=0.2), it will search printers 10 times for every 0.2 seconds for a total of 2 seconds.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];
[sdk open];

[sdk lookupPrinterWithCount:5 interval:1];
```

2-2-4 Cancel To Lookup Printer

It cancels the ongoing printer search.

[Declaration]
-(long) CancelToLookupPrinter;

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL	0XF000	Fail

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];
[sdk open];

// Start Lookup Printer
[sdk lookupPrinterWithCount:10 interval:1];

...

// Cancel Lookup Printer
__SDK_RESULT_CODES_ result = [sdk CancelToLookupPrinter];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-5 Get Paired Devices

It brings up a list of Bluetooth (Classic) devices that are currently paired.

[Declaration]
-(NSArray*) getPairedDevices;

[Return Value]
It returns a list of discovered printers (Bluetooth).

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];
[sdk open];

// XCode 6.3 or Later
//       NSArray<LabelPrinterObject*>* arrPairedDevices = [sdk getPairedDevices];

NSArray* arrPairedDevices = [sdk getPairedDevices];
LabelPrinterObject *printerObj = [arrPairedDevices firstObject];

if (printerObj != nil) {
      NSLog(@"%@", [printerObj getBluetoothDeviceName]);
}
```

2-2-6 Get Searched Devices

It brings up a list of Wi-Fi / Ethernet / BLE devices that are currently discovered.

[Declaration]
-(NSArray*) getSearchedDevices;

[Return Value]
It returns a list of discovered printers (Wi-Fi/Ethernet/BLE).
※ BLE: Bluetooth Low Energy

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];
[sdk open];

// XCoe 6.3 or Later
// NSArray<LabelPrinterObject*>* arrSearchDevices = [sdk getSearchedDevices];

NSArray<LabelPrinterObject*>* arrSearchDevices = [sdk getSearchedDevices];
LabelPrinterObject *printerObj = [arrSearchDevices firstObject];

if (printerObj != nil) {
    NSLog(@"%@", [printerObj getModelName]);
}
```

2-2-7 Connect With Address

This method tries to connect the printer.

[Declaration]
-(long) connectWithAddress:(NSString*)address port:(NSString*) port;

[Parameters]
address: Wi-Fi Printer: Enter the IP Address.
Bluetooth Printer: Enter the Mac Address.
port: Enter the communication port of the printer.(Default : 9100)
Bluetooth printers are not supported.

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL	0XF000	Fail
_SDK_RESULT_FAIL_NO_OPEN	0XF002	SDK is not open
_SDK_RESULT_FAIL_OPEN_ALREADY	0XF003	SDK is already open
_SDK_RESULT_FAIL_CONNECT_ALREADY	0XF005	Printer is already connected
_SDK_RESULT_FAIL_INVALID_PARAMETER	0XF008	Invalid parameter

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];
[sdk open];

NSString *address = @"192.168.0.249";
NSString *port = @"9100";

__SDK_RESULT_CODES_ result = [sdk connectWithAddress:address port:port];

if (result == _SDK_RESULT_SUCCESS) {
    ...
} else {
    ...
}
```

2-2-8 Connect With Target: timeout

This method tries to connect to the printer. (Target Printer Use)

[Declaration]

-(long) connectWithTarget:(LabelPrinterObject*)printer
timeout:(NSInteger)timeout;

[Parameters]

target: It specifies the printer to be connected.
timeout: It specifies the timeout value to use when connecting to the printer.

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL	0XF000	Fail
_SDK_RESULT_FAIL_NO_OPEN	0XF002	SDK is not open
_SDK_RESULT_FAIL_OPEN_ALREADY	0XF003	SDK is already open
_SDK_RESULT_FAIL_CONNECT_ALREADY	0XF005	Printer is already connected
_SDK_RESULT_FAIL_INVALID_PARAMETER	0XF008	Invalid parameter

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

LabelPrinterObject* printer = [[sdk getPairedDevices] firstObject];

if (printer == nil) {
    return;
}

__SDK_RESULT_CODES_ result = [sdk connectWithTarget:printer timeout:10];

if (result == _SDK_RESULT_SUCCESS) {
    ...
} else {
    ...
}
```

2-2-9 Connect With Serial Number

This method tries to connect to the printer. (Serial Number Use)

[Declaration]
-(long) connectWithSerialNumber:(NSString*)serialNumber;

[Parameters]
serialNumber: It specifies the Serial Number of the printer to be connected.
(ex: STD123456789012)
Wi-Fi/Ethernet interface is not supported.

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL	0XF000	Fail
_SDK_RESULT_FAIL_NO_OPEN	0XF002	SDK is not open
_SDK_RESULT_FAIL_OPEN_ALREADY	0XF003	SDK is already open
_SDK_RESULT_FAIL_CONNECT_ALREADY	0XF005	Printer is already connected
_SDK_RESULT_FAIL_INVALID_PARAMETER	0XF008	Invalid parameter

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

NSString *serialNo = @"STD123456789012";

__SDK_RESULT_CODES_ result = [sdk connectWithSerialNumber:serialNo];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-10 Is Connected

This method checks the status of printer connection.

[Declaration]
-(BOOL) isConnected;

[Return Value]
YES: printer is connected.
NO: printer is not connected.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

bool isConnected = [sdk isConnected];

if (isConnected == YES) {
        ...
} else {
        ...
}
```

2-2-11 Disconnect

This method disconnects the printer.

[Declaration]
-(long) disconnect;

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL	0XF000	Fail

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result = [sdk disconnect];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-12 Disconnect With Timeout

This method disconnects the printer within the specified Timeout period.

[Declaration]
-(long) disconnectWithTimeout:(NSInteger)timeout;

[Parameters]
timeout: Set the Timeout period. (Unit: Second)

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL	0XF000	Fail

If data remains in the output buffer,
buffer will be cleared and printer will be disconnected within the Timeout period.
If this parameter is set to -1, this method does not return until all buffers are cleared.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result = [sdk disconnectWithTimeout:10];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-13 Do Print

Let the printer start printing the content of image buffer.

[Declaration]
-(long) doPrint:(NSInteger)numberOfCopies;

[Parameters]
numberOfCopies: Number of label sets

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

if (sdk.isConnected == NO) {
      return;
}

NSInteger copies = 5;

__SDK_RESULT_CODES_ result = [sdk doPrint:copies];

if (result == _SDK_RESULT_SUCCESS ) {
      ...
} else {
      ...
}
```

2-2-14 Set Current Encoding

Sets the encoding to use for text print.

[Declaration]
-(__SDK_RESULT_CODES_) setCurrentEncoding:(NSStringEncoding)encoding

[Parameters]
encoding: Set the encoding type to use.

Code	Value	Description
_SDK_ENCODING_OPTION_AUTOMATIC	0xFFFFFFFF	Auto
_SDK_ENCODING_OPTION_ASCII	0x00000001	ASCII
_SDK_ENCODING_OPTION_KSC5601	0x80000422	Korean graphic character set for information interchange
_SDK_ENCODING_OPTION_SHIFT_JIS	0x00000008	Shift Japanese industrial standards
_SDK_ENCODING_OPTION_BIG5	0x80000a03	Traditional Chinese characters
_SDK_ENCODING_OPTION_GB2312	0x80000421	Simplified Chinese characters(94x94)
_SDK_ENCODING_OPTION_GB18030	0x80000632	Simplified Chinese characters

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result =
      [sdk setCurrentEncoding: _SDK_ENCODING_OPTION_ASCII];

if (result == _SDK_RESULT_SUCCESS) {
      NSLog(@"SUCCESS");
} else {
      NSLog(@"FAIL");
}
```

2-2-15 Draw Text Device Font

Draw text (Device Font) on the image buffer

[Declaration]

```
-(long) drawTextDeviceFont:(NSString*)text
                 xPosition:(NSInteger)xPostion
                 yPosition:(NSInteger)yPostion
            fontSelection:(char)fontSelection
                 fontWidth:(NSInteger)fontWidth
                 fontHeight:(NSInteger)fontHeight
  rightSideCharacterSpacing:(NSInteger)rightSideCharacterSpacing
              fontRotation:(NSInteger)fontRotation
                   reverse:(BOOL)reverse
                      bold:(BOOL)bold
             textAlignment:(NSInteger)textAlignment;
```

[Parameters]

text: Text string to print.

xPosition: Horizontal position (X)
(Reference: 1-3 Available range of X/Y coordinates for each model)

yPosition: Vertical position (Y)
(Reference: 1-3 Available range of X/Y coordinates for each model)

fontSelection: Font selection

Code	Value	Description
_SDK_DEVICE_FONT_6PT	'0'	9 X 15 (dots)
_SDK_DEVICE_FONT_8PT	'1'	12 X 20 (dots)
_SDK_DEVICE_FONT_10PT	'2'	16 X 25 (dots)
_SDK_DEVICE_FONT_12PT	'3'	19 X 30 (dots)
_SDK_DEVICE_FONT_15PT	'4'	24 X 38 (dots)
_SDK_DEVICE_FONT_20PT	'5'	32 X 40 (dots)
_SDK_DEVICE_FONT_30PT	'6'	48 X 76 (dots)
_SDK_DEVICE_FONT_14PT	'7'	22 X 34 (dots)
_SDK_DEVICE_FONT_18PT	'8'	28 X 44 (dots)
_SDK_DEVICE_FONT_24PT	'9'	37 X 58 (dots)
_SDK_DEVICE_FONT_KOREAN1	'a'	16 X 16 (dots) (ASCII 9 X 15)
_SDK_DEVICE_FONT_KOREAN2	'b'	24 X 24 (dots) (ASCII 12 X 24)
_SDK_DEVICE_FONT_KOREAN3	'c'	20 X 20 (dots) (ASCII 12 X 20)
_SDK_DEVICE_FONT_KOREAN4	'd'	26 X 26 (dots) (ASCII 16 X 30)
_SDK_DEVICE_FONT_KOREAN5	'e'	20 X 26 (dots) (ASCII 16 X 30)
_SDK_DEVICE_FONT_KOREAN6	'f'	38 X 38 (dots) (ASCII 22 X 34)
_SDK_DEVICE_FONT_GB2312	'm'	24 X 24 (dots) (ASCII 12 X 24)
_SDK_DEVICE_FONT_BIG5	'n'	24 X 24 (dots) (ASCII 12 X 24)

_SDK_DEVICE_FONT_SHIFT_JIS	'j'	24 X 24 (dots) (ASCII 12 X 24)

fontWidth: Horizontal multiplier (range: 1~4)
fontHeight: Vertical multiplier (range: 1~4)
rightSideCharacterSpacing: Right side character spacing (ex: 5, +3, -10...)
fontRotation: Rotation

Code	Value	Description
_SDK_ROTATION_DEGREES_0	0	No rotation
_SDK_ROTATION_DEGREES_90	1	90 degrees of rotation
_SDK_ROTATION_DEGREES_180	2	180 degrees of rotation
_SDK_ROTATION_DEGREES_270	3	270 degrees of rotation.

reverse: Using Reverse Text (YES: Use, NO: Not use)
bold: Using **Bold Text**(YES: Use, NO: Not use)
textAlignment: Text Alignment

Code	Value	Description
_SDK_ALIGNMENT_LEFT	0	Align to left
_SDK_ALIGNMENT_RIGHT	1	Align to right
_SDK_ALIGNMENT_CENTER	2	Align to center
_SDK_ALIGNMENT_STRING_FROM _RIGHT_2_LEFT	2	Print characters from right to left

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL	0XF000	Fail



- When textAlignment is set to _SDK_ALIGNMENT_CENTER, it has the same effect as the_SDK_ALIGNMENT_STRING_FROM_RIGHT_2_ LEFT option.
- Contents requested by this API will be printed when **2-2-13 doPrint API** is called.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result =
[sdk drawTextDeviceFont:@"Bixolon LabelPrinter DeviceFont Test"
                        xPosition:100
                        yPosition:100
                   fontSelection:_SDK_DEVICE_FONT_KOREAN6
                        fontWidth:1
                       fontHeight:1
rightSideCharacterSpacing:0
                    fontRotation:_SDK_ROTATION_DEGREES_0
                         reverse:NO
                            bold:NO
                   textAlignment:_SDK_ALIGNMENT_CENTER];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-16 Draw Text Vector Font

Draw text (Vector Font) string on the image buffer.

[Declaration]

```
-(long) drawTextVectorFont:(NSString*)text
                  xPosition:(NSInteger)xPostion
                  yPosition:(NSInteger)yPostion
             fontSelection:(char)fontSelection
                  fontWidth:(NSInteger)fontWidth
                 fontHeight:(NSInteger)fontHeight
  rightSideCharacterSpacing:(NSInteger)rightSideCharacterSpacing
               fontRotation:(NSInteger)fontRotation
                    reverse:(BOOL)reverse
                       bold:(BOOL)bold
                     italic:(BOOL)italic
  textWriteDirectionRightToLeft:(BOOL)textWriteDirectionRightToLeft
                  textAlignment:(NSInteger)textAlignment;
```

[Parameters]

text: Text string to print

xPosition: Horizontal position (X)
(Reference: 1-3 Available range of X/Y coordinates for each model)

yPosition: Vertical position (Y)
(Reference: 1-3 Available range of X/Y coordinates for each model)

fontSelection: Font selection

Code	Value	Description
_SDK_VECTOR_FONT_ASCII	'U'	ASCII (1Byte code)
_SDK_VECTOR_FONT_KS5601	'K'	KS5601 (2Byte code)
_SDK_VECTOR_FONT_BIG5	'B'	BIG5 (2Byte code)
_SDK_VECTOR_FONT_GB2312	'G'	GB2312 (2Byte code)
_SDK_VECTOR_FONT_SHIFT_JIS	'J'	Shift-JIS (2Byte code)
_SDK_VECTOR_FONT_OCR_A	'a'	OCR-A (1Byte code)
_SDK_VECTOR_FONT_OCR_B	'b'	OCR-B (1Byte code)

fontWidth: Font Width (range: 1~1500).
fontHeight: Font height (range: 1~1500).
rightSideCharacterSpacing: Right-side character spacing (ex: 5, +3, -10...).
fontRotation: Rotation

Code	Value	Description
_SDK_ROTATION_DEGREES_0	0	No rotation
_SDK_ROTATION_DEGREES_90	1	90 degrees of rotation
_SDK_ROTATION_DEGREES_180	2	180 degrees of rotation
_SDK_ROTATION_DEGREES_270	3	270 degrees of rotation.

reverse: Using Reverse Text (YES: Use, NO: Not use)
bold: Using **Bold Text**(YES: Use, NO: Not use)
italic: Using italic (YES: Use, NO: Not use)
textWriteDirectionRightToLeft: Text string write direction(YES: right to left, NO: left to right)
textAlignment: Text Alignment

Code	Value	Description
_SDK_ALIGNMENT_LEFT	0	Align to left
_SDK_ALIGNMENT_RIGHT	1	Align to right
_SDK_ALIGNMENT_CENTER	2	Align to center
_SDK_ALIGNMENT_STRING_FROM_RIGHT_2_LEFT	2	Print characters from right to left

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL	0XF000	Fail

- When textAlignment is set to _SDK_ALIGNMENT_STRING_FROM _RIGHT_2_LEFT, it has the same effect as the _SDK_ALIGNMENT_ CENTER option.
- Contents requested by this API will be printed when **2-2-13 doPrint API** is called.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result =
[sdk drawTextVectorFont:@"Bixolon LabelPrinter VectorFont Test"
                                xPosition:100
                                yPosition:100
                           fontSelection:_SDK_VECTOR_FONT_KS5601
                                fontWidth:30
                               fontHeight:30
        rightSideCharacterSpacing:0
                            fontRotation:_SDK_ROTATION_DEGREES_0
                                 reverse:NO
                                    bold:NO
                                  italic:NO
  textWriteDirectionRightToLeft:NO
                           textAlignment:_SDK_ALIGNMENT_CENTER
];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-17 Draw Barcode1D

Draw 1D Barcode on the image buffer.

[Declaration]

```
-(long) drawBarcode1D:(NSString*)data
                xPosition:(NSInteger)xPostion
                yPosition:(NSInteger)yPostion
              barcodeType:(NSInteger)barcodeType
             widthNarrow:(NSInteger)widthNarrow
               widthWide:(NSInteger)widthWide
                  height:(NSInteger)height
                     hri:(NSInteger)hri
          quietZoneWidth:(NSInteger)quietZoneWidth
                rotation:(NSInteger)rotation;
```

[Parameters]

data: Barcode data to print
xPosition: Horizontal position (X)
(Reference: 1-3 Available range of X/Y coordinates for each model)
yPosition: Vertical position (Y)
(Reference: 1-3 Available range of X/Y coordinates for each model)
barcodeType: Barcode selection

Code	Value	Description
_SDK_BARCODE_TYPE_CODE39	0	Code39
_SDK_BARCODE_TYPE_CODE128	1	Code128
_SDK_BARCODE_TYPE_I2Of5	2	Interleaved 2of5
_SDK_BARCODE_TYPE_CODABAR	3	Codabar
_SDK_BARCODE_TYPE_CODE93	4	Code93
_SDK_BARCODE_TYPE_UPC_A	5	UPC-A
_SDK_BARCODE_TYPE_UPC_E	6	UPC-E
_SDK_BARCODE_TYPE_EAN13	7	EAN13
_SDK_BARCODE_TYPE_EAN8	8	EAN8
_SDK_BARCODE_TYPE_EAN128	9	UCC/EAN128
_SDK_BARCODE_TYPE_CODE11	10	Code11
_SDK_BARCODE_TYPE_PLANET	11	Planet
_SDK_BARCODE_TYPE_INDUSTRIAL_2Of5	12	Industrial 2of5
_SDK_BARCODE_TYPE_STANDARD_2Of5	13	Standard 2of5
_SDK_BARCODE_TYPE_LOGMARS	14	Logmars
_SDK_BARCODE_TYPE_UPC_EAN_EXTENSIONS	15	UPC/EAN Extensions
_SDK_BARCODE_TYPE_POSTNET	16	Postnet

widthNarrow: Narrow bar width
widthWide: Wide bar width
height: Barcode height

hri: HRI (Human Readable Interpretation)

Code	Value	Description
_SDK_BARCODE_HRI_NONE	0	HRI is not used.
_SDK_BARCODE_HRI_BELOW_FONTSIZE1	1	Position of HRI: Below barcode Font Size: 1
_SDK_BARCODE_HRI_ABOVE_FONTSIZE1	2	Position of HRI: Above barcode Font Size: 1
_SDK_BARCODE_HRI_BELOW_FONTSIZE2	3	Position of HRI: Below barcode Font Size: 2
_SDK_BARCODE_HRI_ABOVE_FONTSIZE2	4	Position of HRI: Above barcode Font Size: 2
_SDK_BARCODE_HRI_BELOW_FONTSIZE3	5	Position of HRI: Below barcode Font Size: 3
_SDK_BARCODE_HRI_ABOVE_FONTSIZE3	6	Position of HRI: Above barcode Font Size: 3
_SDK_BARCODE_HRI_BELOW_FONTSIZE4	7	Position of HRI: Below barcode Font Size: 4
_SDK_BARCODE_HRI_ABOVE_FONTSIZE4	8	Position of HRI: Above barcode Font Size: 4

quietZoneWidth: Quiet zone is added to the front and end of the barcode for safe scanning.(0~20)

rotation: Rotation

Code	Value	Description
_SDK_ROTATION_DEGREES_0	0	No rotation
_SDK_ROTATION_DEGREES_90	1	90 degrees of rotation
_SDK_ROTATION_DEGREES_180	2	180 degrees of rotation
_SDK_ROTATION_DEGREES_270	3	270 degrees of rotation.

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL	0XF000	Fail
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data
_SDK_RESULT_FAIL_INVALID_PARAMETER	0XF008	Invalid parameter

Contents requested by this API will be printed when **2-2-13 doPrint API** is called.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result =
[sdk drawBarcode1D:@"1234567890128"
             xPosition:100
             yPosition:100
          barcodeType:_SDK_BARCODE_TYPE_UPC_A
          widthNarrow:2
             widthWide:2
                 height:100
                     hri:_SDK_BARCODE_HRI_BELOW_FONTSIZE2
     quietZoneWidth:0
              rotation:_SDK_ROTATION_DEGREES_0];

if (result == _SDK_RESULT_SUCCESS) {
       ...
} else {
       ...
}
```

2-2-18 Draw Barcode Maxi Code

Draw Maxicode Barcode on the image buffer.

[Declaration]

```
-(long) drawBarcodeMaxiCode:(NSString*)data
                  xPosition:(NSInteger)xPostion
                  yPosition:(NSInteger)yPostion
                       mode:(NSInteger)mode;
```

[Parameters]

data: Barcode data to print
xPosition: Horizontal position(X)
(Reference: 1-3 Available range of X/Y coordinates for each model)
yPosition: Vertical position(Y)
(Reference: 1-3 Available range of X/Y coordinates for each model)
mode: MaxiCode Mode

Code	Value	Description
_SDK_MAXICODE_MODE_0	0	MaxiCode Mode 0
_SDK_MAXICODE_MODE_2	2	MaxiCode Mode 2
_SDK_MAXICODE_MODE_3	3	MaxiCode Mode 3
_SDK_MAXICODE_MODE_4	4	MaxiCode Mode 4

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

Contents requested by this API will be printed when **2-2-13 doPrint API** is called.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...
__SDK_RESULT_CODES_ result =
[sdk drawBarcodeMexiCode:@"Bixolon Maxicode Test"
      xPosition:100
      yPosition:100
            mode:_SDK_MEXICODE_MODE_0];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-19 Draw Barcode PDF417

Draw PDF417 Barcode on the image buffer.

[Declaration]

```
-(long) drawBarcodePDF417:(NSString*)data
                 xPosition:(NSInteger)xPostion
                 yPosition:(NSInteger)yPostion
          maximumRowCount:(NSInteger)maximumRowCount
       maximumColumnCount:(NSInteger)maximumColumnCount
     errorCorrectionLevel:(NSInteger)errorCorrectionLevel
   dataCompressionMethod:(NSInteger)dataCompressionMethod
        printBarcodeText:(BOOL)printBarcodeText
     barcodeOriginPoint:(NSInteger)barcodeOriginPoint
             moduleWidth:(NSInteger)moduleWidth
               barHeight:(NSInteger)barHeight
                rotation:(NSInteger)rotation;
```

[Parameters]

data: Barcode data to print
xPosition: Horizontal position (X)
(Reference: 1-3 Available range of X/Y coordinates for each model)
yPosition: Vertical position (Y)
(Reference: 1-3 Available range of X/Y coordinates for each model)
maximumRowCount: Maximum Row Count (range: 3~90)
maximumColumnCount: Maximum Column Count (range: 1 ~ 30)
errorCorrectionLevel: Error correction level

Code	Value	Description
_SDK_ERROR_CORRECTION_LEVEL0	0	Error Correction Level 0
_SDK_ERROR_CORRECTION_LEVEL1	1	Error Correction Level 1
_SDK_ERROR_CORRECTION_LEVEL2	2	Error Correction Level 2
_SDK_ERROR_CORRECTION_LEVEL3	3	Error Correction Level 3
_SDK_ERROR_CORRECTION_LEVEL4	4	Error Correction Level 4
_SDK_ERROR_CORRECTION_LEVEL5	5	Error Correction Level 5
_SDK_ERROR_CORRECTION_LEVEL6	6	Error Correction Level 6
_SDK_ERROR_CORRECTION_LEVEL7	7	Error Correction Level 7
_SDK_ERROR_CORRECTION_LEVEL8	8	Error Correction Level 8

dataCompressionMethod: Data compression method

Code	Value	Description
_SDK_DATA_COMPRESSION_METHOD_TEXT	0	2char / codeword
_SDK_DATA_COMPRESSION_METHOD_NUMERIC	1	2.93 char / codeword
_SDK_DATA_COMPRESSION_METHOD_BINARY	2	1.2bytes / codeword

printBarcodeText: HRI (YES: below the barcode, NO: Not printed)
barcodeOriginPoint: Barcode origin point

Code	Value	Description
_SDK_BARCODE_ORIGIN_POINT_CENTER	0	Set the reference point of the barcode to the center.
_SDK_BARCODE_ORIGIN_POINT_UpperLeft	1	Set the reference point of the barcode in the upper left corner.

moduleWidth: Module Width (range: 2~9)
BarHeight: Bar Height (range: 4~99)
rotation: Rotation

Code	Value	Description
_SDK_ROTATION_DEGREES_0	0	No rotation
_SDK_ROTATION_DEGREES_90	1	90 degrees of rotation
_SDK_ROTATION_DEGREES_180	2	180 degrees of rotation
_SDK_ROTATION_DEGREES_270	3	270 degrees of rotation.

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data



Contents requested by this API will be printed when **2-2-13 doPrint API** is called.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result =
[sdk drawBarcodePDF417:@"Bixolon PDF417 Test!!!"
                   xPosition:100
                   yPosition:100
          maximumRowCount:10
      maximumColumnCount:5
  errorCorrectionLevel:_SDK_ERROR_CORRECTION_LEVEL0
dataCompressionMethod:_SDK_DATA_COMPRESSION_METHOD_TEXT
         printBarcodeText:false
     barcodeOriginPoint:_SDK_BARCODE_ORIGIN_POINT_CENTER
                moduleWidth:5
                   barHeight:50
                     rotation:_SDK_ROTATION_DEGREES_0];

if (result == _SDK_RESULT_SUCCESS) {
       ...
} else {
       ...
}
```

2-2-20 Draw Barcode QR Code

Draw QRCode Barcode on the image buffer.

[Declaration]

```
-(long) drawBarcodeQRCode:(NSString*)data
                 xPosition:(NSInteger)xPostion
                 yPosition:(NSInteger)yPostion
               barcodeSize:(NSInteger)barcodeSize
                     model:(NSInteger)model
        errorColectionLevel:(NSInteger)errorColectionLevel
                  rotation:(NSInteger)rotation;
```

[Parameters]

data: Barcode data to print

xPosition: Horizontal position (X)
(Reference: 1-3 Available range of X/Y coordinates for each model)

yPosition: Vertical position (Y)
(Reference: 1-3 Available range of X/Y coordinates for each model)

barcodeSize: Barcode size (range: 1~9)

model: Model Selection

Code	Value	Description
_SDK_QRCODE_MODEL_1	1	QR Model 1
_SDK_QRCODE_MODEL_2	2	QR Model 2

errorCorrectionLevel: ECC Level

Value	Description
0	7%
1	15%
2	25%
3	30%

rotation: Rotation

Code	Value	Description
_SDK_ROTATION_DEGREES_0	0	No rotation
_SDK_ROTATION_DEGREES_90	1	90 degrees of rotation
_SDK_ROTATION_DEGREES_180	2	180 degrees of rotation
_SDK_ROTATION_DEGREES_270	3	270 degrees of rotation.

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

Contents requested by this API will be printed when **2-2-13 doPrint API** is called.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

if (sdk.isConnected == NO) {
      return;
}

__SDK_RESULT_CODES_ result =
[sdk drawBarcodeQRCode:@"Bixolon QRCode Test!!!"
                     xPosition:50
                     yPosition:50
                  barcodeSize:2
                        model:_SDK_QRCODE_MODEL_1
      errorColectionLevel:_SDK_ERROR_CORRECTION_LEVEL0
                   rotation:_SDK_ROTATION_DEGREES_0];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-21 Draw Barcode Data Matrix

Draw QRCode Barcode on the image buffer.

[Declaration]

```
-(long) drawBarcodeDataMatrix:(NSString*)data
                    xPosition:(NSInteger)xPostion
                    yPosition:(NSInteger)yPostion
                   barcodeSize:(NSInteger)barcodeSize
                      reverse:(BOOL)reverse
                     rotation:(NSInteger)rotation;
```

[Parameters]

data: Barcode data to print
xPosition: Horizontal position (X)
(Reference: 1-3 Available range of X/Y coordinates for each model)
yPosition: Vertical position (Y)
(Reference: 1-3 Available range of X/Y coordinates for each model)
barcodeSize: Barcode Size (range: 1~9)
reverse: Reverse mode (YES: Reverse, NO: Normal)
rotation: Rotation

Code	Value	Description
_SDK_ROTATION_DEGREES_0	0	No rotation
_SDK_ROTATION_DEGREES_90	1	90 degrees of rotation
_SDK_ROTATION_DEGREES_180	2	180 degrees of rotation
_SDK_ROTATION_DEGREES_270	3	270 degrees of rotation.

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

Contents requested by this API will be printed when **2-2-13 doPrint API** is called.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result =
[sdk drawBarcodeDataMatrix:@"Bixolon DataMatrix Test!!!"
                  xPosition:50
                  yPosition:50
                barcodeSize:2
                    reverse:NO
                   rotation:_SDK_ROTATION_DEGREES_0];

if (result == _SDK_RESULT_SUCCESS) {
    ...
} else {
    ...
}
```

2-2-22 Draw Barcode Aztec

Draw Aztec Barcode on the image buffer.

[Declaration]

```
-(long) drawBarcodeAztec:(NSString*)data
                xPosition:(NSInteger)xPostion
                yPosition:(NSInteger)yPostion
              barcodeSize:(NSInteger)barcodeSize
          extendedChannel:(BOOL)extendedChannel
       errorCorrectionLevel:(NSInteger)errorCorrectionLevel
                menuSymbol:(BOOL)menuSymbol
          numberOfSymbols:(NSInteger)numberOfSymbols
                optionalID:(NSString*)optionalID
                  rotation:(NSInteger)rotation;
```

[Parameters]

data: Barcode data to print
xPosition: Horizontal position (X)
(Reference: 1-3 Available range of X/Y coordinates for each model)
yPosition: Vertical position (Y)
(Reference: 1-3 Available range of X/Y coordinates for each model)
barcodeSize: Barcode size (range: 1~10)
extendedChannel: Extended channel interpretation code (YES: Enable, NO: Disable)
errorCorrectionLevel: Error control and symbol size/type

Value	Error control and symbol size/type
0	Default error correction level
1 ~ 99	Error correction percentage

menuSymbol: Menu symbol (YES: Enable, NO: Disable)
numberOfSymbols: Number of symbols for structured append: (range: 1 ~ 26)
optionalID: Optional ID field for structured append: ID field string
(Maximum 24 character)
rotation: Rotation

Code	Value	Description
_SDK_ROTATION_DEGREES_0	0	No rotation
_SDK_ROTATION_DEGREES_90	1	90 degrees of rotation
_SDK_ROTATION_DEGREES_180	2	180 degrees of rotation
_SDK_ROTATION_DEGREES_270	3	270 degrees of rotation.

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

Contents requested by this API will be printed when **2-2-13 doPrint API** is called.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result =
[sdk drawBarcodeAztec:@"Bixolon Aztec Test!!!"
                xPosition:50
                yPosition:50
            barcodeSize:5
      extendedChannel:false
errorCorrectionLevel:0
              menuSymbol:false
      numberOfSymbols:10
              optionalID:nil
                rotation:_SDK_ROTATION_DEGREES_0];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-23 Draw Barcode Code49

Draw Code49 Barcode on the image buffer.

[Declaration]

```
-(long)drawBarcodeCode49:(NSString*)data
                xPosition:(NSInteger)xPostion
                yPosition:(NSInteger)yPostion
              widthNarrow:(NSInteger)widthNarrow
                widthWide:(NSInteger)widthWide
                   height:(NSInteger)height
                      hri:(NSInteger)hri
            startingMode:(NSInteger)startingMode
                 rotation:(NSInteger)rotation;
```

[Parameters]

data: Barcode data to print
xPosition: Horizontal position (X)
(Reference: 1-3 Available range of X/Y coordinates for each model)
yPosition: Vertical position (Y)
(Reference: 1-3 Available range of X/Y coordinates for each model)
widthNarrow: Narrow bar width
widthWide: Wide bar width
hri: HRI

Value	Description
0	Not Printed
1	Below the barcode
2	Above the barcode

height: Barcode height
startingMode: Starting mode

Code	Value	Description
_SDK_STARTINGMODE_REGULAR_ALPHANUMERIC	0	Regular Alphanumeric Mode
_SDK_STARTINGMODE_MULTIPLE_READ_ALPHANUMERIC	1	Multiple Read Alphanumeric
_SDK_STARTINGMODE_REGULAR_NUMERIC	2	Regular Numeric Mode
_SDK_STARTINGMODE_GROUP_ALPHANUMERIC	3	Group Alphanumeric Mode
_SDK_STARTINGMODE_REGULAR_ALPHANUMRIC_SHIFT1	4	Regular Alphanumeric Shift 1
_SDK_STARTINGMODE_REGULAR_ALPHANUMRIC_SHIFT2	5	Regular Alphanumeric Shift 2
_SDK_STARTINGMODE_AUTOMATIC	7	Automatic Mode

rotation: Rotation

Code	Value	Description
_SDK_ROTATION_DEGREES_0	0	No rotation
_SDK_ROTATION_DEGREES_90	1	90 degrees of rotation
_SDK_ROTATION_DEGREES_180	2	180 degrees of rotation
_SDK_ROTATION_DEGREES_270	3	270 degrees of rotation.

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data



Contents requested by this API will be printed when **2-2-13 doPrint API** is called.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result =
[sdk drawBarcodeCode49:@"Bixolon Code49 Test!!!"
                 xPosition:50
                 yPosition:50
               widthNarrow:4
                 widthWide:2
                    height:100
                       hri:_SDK_BARCODE_HRI_BELOW_FONTSIZE1
             startingMode:_SDK_STARTINGMODE_AUTOMATIC
                  rotation:_SDK_ROTATION_DEGREES_0];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-24 Draw Barcode Coda Block

Draw CodaBlock Barcode on the image buffer.

[Declaration]

```
-(long) drawBarcodeCodaBlock:(NSString*)data
                   xPosition:(NSInteger)xPostion
                   yPosition:(NSInteger)yPostion
                 widthNarrow:(NSInteger)widthNarrow
                   widthWide:(NSInteger)widthWide
                      height:(NSInteger)height
               securityLevel:(BOOL)securityLevel
       numberOfCharactersPerrow:(NSInteger)numberOfCharactersPerrow
                        mode:(char)mode
          numberOfRowToEncode:(NSInteger)numberOfRowToEncode;
```

[Parameters]

data: Barcode data to print
xPosition: Horizontal position (X)
(Reference: 1-3 Available range of X/Y coordinates for each model)
yPosition: Vertical position (Y)
(Reference: 1-3 Available range of X/Y coordinates for each model)
widthNarrow: Narrow bar width
widthWide: Wide bar width
height: Barcode height
securityLevel: Security level (YES: Enable, NO: Disable)
numberOfCharactersPerrow: Number of characters per row (data columns): 2~62
mode: Mode

Code	Value	Description
_SDK_CODABLACK_MODE_A	'A'	Code 39 character set is used.
_SDK_CODABLACK_MODE_E	'E'	Code 128 character set is used.
_SDK_CODABLACK_MODE_F	'F'	Code 128 character set is used. Function 1 is added automatically.

numberOfRowToEncode: Number of rows to encode

Mode	Value
_SDK_CODABLACK_MODE_A	1 ~ 18
_SDK_CODABLACK_MODE_E	2 ~ 4
_SDK_CODABLACK_MODE_F	2 ~ 4

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

Contents requested by this API will be printed when **2-2-13 doPrint API** is called.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result =
[sdk drawBarcodeCodaBlock:@"Bixolon CodaBlock Test!!!"
                 xPosition:100
                 yPosition:100
               widthNarrow:4
                  widthWide:2
                      height:100
             securityLevel:false
   numberOfCharactersPerrow:10
                      mode:_SDK_CODABLACK_MODE_A
          numberOfRowToEncode:10];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-25 Draw Barcode Micro PDF

Draw Micro PDF 417 Barcode on the image buffer.

[Declaration]

```
-(long) drawBarcodeMicroPDF:(NSString*)data
                  xPosition:(NSInteger)xPostion
                  yPosition:(NSInteger)yPostion
                moduleWidth:(NSInteger)moduleWidth
             barcodeHeight:(NSInteger)barcodeHeight
                       mode:(NSInteger)mode
                   rotation:(NSInteger)rotation;
```

[Parameters]

text: Barcode data to print
xPosition: Horizontal position (X)
(Reference: *1-3 Available range of X/Y coordinates for each model*)
yPosition: Vertical position (Y)
(Reference: *1-3 Available range of X/Y coordinates for each model*)
moduleWidth: Module Width (range: 2~8)
barcodeHeight: Barcode height (range: 1~99)
mode: mode

Mode (M)	Number of Data Columns	Number of Data Rows	% of Cws for EC	Max Alpha Characters	Max Digits	Remarks
0	1	11	64	6	8	-
1	1	14	50	12	17	-
2	1	17	41	18	26	-
3	1	20	40	22	32	-
4	1	24	33	30	44	-
5	1	28	29	38	55	-
6	2	8	50	14	20	-
7	2	11	41	24	35	-
8	2	14	32	36	52	-
9	2	17	29	46	67	-
10	2	20	28	56	82	-
11	2	23	28	64	93	-
12	2	26	29	72	105	-
13	3	6	67	10	14	-
14	3	8	58	18	26	-
15	3	10	53	26	38	-
16	3	12	50	34	49	-
17	3	15	47	46	67	-

18	3	20	43	66	96	-
19	3	26	41	90	132	-
20	3	32	40	114	167	-
21	3	38	39	138	202	-
22	3	44	38	162	237	-
23	4	6	50	22	32	-
24	4	8	44	34	49	-
25	4	10	40	46	67	-
26	4	12	38	58	85	-
27	4	15	35	76	111	-
28	4	20	33	106	155	-
29	4	26	31	142	208	-
30	4	32	30	178	261	-
31	4	38	29	214	313	-
32	4	44	28	250	366	-
33	4	4	50	14	20	-

rotation: Rotation

Code	Value	Description
_SDK_ROTATION_DEGREES_0	0	No rotation
_SDK_ROTATION_DEGREES_90	1	90 degrees of rotation
_SDK_ROTATION_DEGREES_180	2	180 degrees of rotation
_SDK_ROTATION_DEGREES_270	3	270 degrees of rotation.

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

Contents requested by this API will be printed when **2-2-13 doPrint API** is called.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result =
[sdk drawBarcodeMicroPDF:@"Bixolon MicroPDF417 Test!!!"
                       xPosition:100
                       yPosition:100
                     moduleWidth:4
                  barcodeHeight:100
                           mode:0
                       rotation:_SDK_ROTATION_DEGREES_0];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-26 Draw Barcode IMB

Draw IMB(Inteligent Mail Barcode) on the image buffer.

[Declaration]

```
-(long) drawBarcodeIMB:(NSString*)data
                xPosition:(NSInteger)xPostion
                yPosition:(NSInteger)yPostion
          printBarcodeText:(BOOL)printBarcodeText
                rotation:(NSInteger)rotation;
```

[Parameters]

data: Barcode data to print
xPosition: Horizontal position (X)
(Reference: 1-3 Available range of X/Y coordinates for each model)
yPosition: Vertical position (Y)
(Reference: 1-3 Available range of X/Y coordinates for each model)
printBarcodeText: HRI (YES: bellow the barcode, NO: Not printed)
rotation: Rotation

Code	Value	Description
_SDK_ROTATION_DEGREES_0	0	No rotation
_SDK_ROTATION_DEGREES_90	1	90 degrees of rotation
_SDK_ROTATION_DEGREES_180	2	180 degrees of rotation
_SDK_ROTATION_DEGREES_270	3	270 degrees of rotation.

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

Contents requested by this API will be printed when **2-2-13 doPrint API** is called.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...
__SDK_RESULT_CODES_ result =
[sdk drawBarcodeIMB:@"Bixolon Inteligent Mail Barcode Test!!!"
              xPosition:100
              yPosition:100
   printBarcodeText:NO
               rotation:_SDK_ROTATION_DEGREES_0];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-27 Draw Barcode MSI

Draw MSI Barcode on the image buffer.

[Declaration]

```
-(long) drawBarcodeMSI:(NSString*)data
                xPosition:(NSInteger)xPostion
                yPosition:(NSInteger)yPostion
              widthNarrow:(NSInteger)widthNarrow
                widthWide:(NSInteger)widthWide
                   height:(NSInteger)height
        checkDigitSelection:(NSInteger)checkDigitSelection
      printCheckDigitInHRI:(BOOL)printCheckDigitInHRI
                      hri:(NSInteger)hri
                 rotation:(NSInteger)rotation;
```

[Parameters]

data: Barcode data to print
xPosition: Horizontal position (X)
(Reference: 1-3 Available range of X/Y coordinates for each model)
yPosition: Vertical position (Y)
(Reference: 1-3 Available range of X/Y coordinates for each model)
widthNarrow: Narrow bar width
widthWide: Wide bar width
height: Barcode height
checkDigitSelection: Check digit selection

Code	Value	Description
_SDK_CHECKDIGIT_NONE	0	No Check Digit
_SDK_CHECKDIGIT_1MOD10	1	Check Digit 1 Mod 10
_SDK_CHECKDIGIT_2MOD10	2	Check Digit 2 Mod 10
_SDK_CHECKDIGIT_1MOD11_AND_1MOD_10	3	Check Digit 1 Mod 10

printCheckDigitInHRI: Print check digit in HRI (YES: Print, NO: Not print)
hri: HRI

Value	Description
0	Not printed
1	Below the bar code
2	Above the bar code

rotation: Rotation

Code	Value	Description
_SDK_ROTATION_DEGREES_0	0	No rotation
_SDK_ROTATION_DEGREES_90	1	90 degrees of rotation
_SDK_ROTATION_DEGREES_180	2	180 degrees of rotation
_SDK_ROTATION_DEGREES_270	3	270 degrees of rotation.

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data



Contents requested by this API will be printed when **2-2-13 doPrint API** is called.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result =
[sdk drawBarcodeMSI:@"943457842"
              xPosition:100
              yPosition:100
            widthNarrow:4
              widthWide:2
                 height:100
checkDigitSelection:_SDK_CHECKDIGIT_1MOD10
printCheckDigitInHRI:YES
                    hri:_SDK_BARCODE_HRI_BELOW_FONTSIZE1
               rotation:_SDK_ROTATION_DEGREES_0];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-28 Draw Barcode Plessey

Draw Plessey Barcode on the image buffer.

[Declaration]

```
-(long) drawBarcodePlessey:(NSString*)data
                  xPosition:(NSInteger)xPostion
                  yPosition:(NSInteger)yPostion
                widthNarrow:(NSInteger)widthNarrow
                  widthWide:(NSInteger)widthWide
                     height:(NSInteger)height
           printCheckDigit:(BOOL)printCheckDigit
                       hri:(NSInteger)hri
                   rotation:(NSInteger)rotation;
```

[Parameters]

data: Barcode data to print
xPosition: Horizontal position (X)
(Reference: 1-3 Available range of X/Y coordinates for each model)
yPosition: Vertical position (Y)
(Reference: 1-3 Available range of X/Y coordinates for each model)
widthNarrow: Narrow bar width
widthWide: Wide bar width
height: Barcode height
printCheckDigitInHRI: Print check digit
(YES: Print check digit, NO: Not print check digit)
hri: HRI

Value	Description
0	Not printed
1	Below the bar code
2	Above the bar code

rotation: Rotation

Code	Value	Description
_SDK_ROTATION_DEGREES_0	0	No rotation
_SDK_ROTATION_DEGREES_90	1	90 degrees of rotation
_SDK_ROTATION_DEGREES_180	2	180 degrees of rotation
_SDK_ROTATION_DEGREES_270	3	270 degrees of rotation.

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

Contents requested by this API will be printed when **2-2-13 doPrint API** is called.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result =
[sdk drawBarcodePlessey:@"12345678"
                xPosition:100
                yPosition:100
              widthNarrow:4
                widthWide:2
                   height:100
          printCheckDigit:YES
                      hri:_SDK_BARCODE_HRI_BELOW_FONTSIZE1
                 rotation:_SDK_ROTATION_DEGREES_0];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-29 Draw Barcode TLC39

Draw TLC39 Barcode on the image buffer.

[Declaration]

```
-(long) drawBarcodeTLC39:(NSString*)data
                 xPosition:(NSInteger)xPostion
                 yPosition:(NSInteger)yPostion
               widthNarrow:(NSInteger)widthNarrow
                 widthWide:(NSInteger)widthWide
                    height:(NSInteger)height
    rowHeightOfMicroPDF417:(NSInteger)rowHeightOfMicroPDF417
narrowWidthOfMicroPDF417:(NSInteger)narrowWidthOfMicroPDF417
                  rotation:(NSInteger)rotation;
```

[Parameters]

data: Barcode data to print
xPosition: Horizontal position (X)
(Reference: 1-3 Available range of X/Y coordinates for each model)
yPosition: Vertical position (Y)
(Reference: 1-3 Available range of X/Y coordinates for each model)
widthNarrow: Narrow bar width of the Code 39
widthWide: Wide bar width of the Code 39
height: Height of the code39
rowHeightOfMicroPDF417: Row height of microPDF417 (range: 1 ~ 255)
narrowWidthOfMicroPDF417: narrowWideHeight of microPDF417 (range 1 ~ 10)
rotation: Rotation

Code	Value	Description
_SDK_ROTATION_DEGREES_0	0	No rotation
_SDK_ROTATION_DEGREES_90	1	90 degrees of rotation
_SDK_ROTATION_DEGREES_180	2	180 degrees of rotation
_SDK_ROTATION_DEGREES_270	3	270 degrees of rotation.

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

Contents requested by this API will be printed when **2-2-13 doPrint API** is called.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result =
[sdk drawBarcodeTLC39:@"123456,ABCD12345678901234"
                        xPosition:100
                        yPosition:100
                      widthNarrow:4
                        widthWide:2
                           height:100
   rowHeightOfMicroPDF417:3
narrowWidthOfMicroPDF417:2
                         rotation:_SDK_ROTATION_DEGREES_0];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-30 Draw Barcode RSS

Draw RSS Barcode on the image buffer.

[Declaration]

```
-(long) drawBarcodeRSS:(NSString*)data
                 xPosition:(NSInteger)xPostion
                 yPosition:(NSInteger)yPostion
               barcodeType:(NSInteger)barcodeType
             magnification:(NSInteger)magnification
           separatorHeight:(NSInteger)separatorHeight
             barcodeHeight:(NSInteger)barcodeHeight
              segmentWidth:(NSInteger)segmentWidth
                  rotation:(NSInteger)rotation;
```

[Parameters]

data: Barcode data to print

xPosition: Horizontal position (X)
(Reference: 1-3 Available range of X/Y coordinates for each model)

yPosition: Vertical position (Y)
(Reference: 1-3 Available range of X/Y coordinates for each model)

barcodeType: RSS barcode type

Code	Value	Description
_SDK_RSS_BARCODE_TYPE_RSS14	0	RSS14
_SDK_RSS_BARCODE_TYPE_RSS14_TRUNCATED	1	RSS14 truncated
_SDK_RSS_BARCODE_TYPE_RSS14_STACKED	2	RSS14 stacked
_SDK_RSS_BARCODE_TYPE_RSS14_STACKED_OMNIDIRECTIONAL	3	RSS14 Stacked omnidirectional
_SDK_RSS_BARCODE_TYPE_RSS_LIMITED	4	RSS limited
_SDK_RSS_BARCODE_TYPE_RSS_EXPANDED	5	RSS Expanded
_SDK_RSS_BARCODE_TYPE_UPC_A	6	RSS UPC A
_SDK_RSS_BARCODE_TYPE_UPC_E	7	RSS UPC E
_SDK_RSS_BARCODE_TYPE_EAN13	8	EAN13
_SDK_RSS_BARCODE_TYPE_EAN8	9	EAN 8
_SDK_RSS_BARCODE_TYPE_UCC_EAN128_CC_A_B	10	EAN128 CC-A/B
_SDK_RSS_BARCODE_TYPE_UCC_EAN128_CC_C	11	EAN128 CC-C

magnification: magnification. *(range: 1~10)*

separatorHeight: Height of separator (*range:* 1~2)

barcodeHeight: this parameter only affects the UCC/EAN barcode type and CC-A/B/C barcode type.

segmentWidth: This parameter only affects the RSS expanded barcode type (*range:* 0~22 Even number only)

rotation: Rotation

Code	Value	Description
_SDK_ROTATION_DEGREES_0	0	No rotation
_SDK_ROTATION_DEGREES_90	1	90 degrees of rotation
_SDK_ROTATION_DEGREES_180	2	180 degrees of rotation
_SDK_ROTATION_DEGREES_270	3	270 degrees of rotation.

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data



Contents requested by this API will be printed when **2-2-13 doPrint API** is called.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result =
[sdk drawBarcodeRSS:@"12345678901|this is composite info"
              xPosition:100
              yPosition:100
          barcodeType:_SDK_RSS_BARCODE_TYPE_RSS14
        magnification:2
    separatorHeight:1
        barcodeHeight:20
         segmentWidth:10
               rotation:_SDK_ROTATION_DEGREES_0];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-31 Draw Barcode GS1 Data Bar

Draw GS1 DataBar Barcode on the image buffer.

[Declaration]

```
-(__SDK_RESULT_CODES_) drawBarcodeGS1DataBar:(NSString*)data
                                    xPosition:(NSInteger)xPostion
                                    yPosition:(NSInteger)yPostion
                                  barcodeType:(NSInteger)barcodeType
                                 magnification:(NSInteger)magnification
                               separatorHeight:(NSInteger)separatorHeight
                                barcodeHeight:(NSInteger)barcodeHeight
                                segmentWidth:(NSInteger)segmentWidth
                                     rotation:(NSInteger)rotation
```

[Parameters]

data: Barcode data to print

xPosition: Horizontal position (X)
(Reference: 1-3 Available range of X/Y coordinates for each model)

yPosition: Vertical position (Y)
(Reference: 1-3 Available range of X/Y coordinates for each model)

barcodeType: RSS barcode type

Code	Value	Description
_SDK_RSS_BARCODE_TYPE_RSS14	0	RSS14
_SDK_RSS_BARCODE_TYPE_RSS14_TRUNCATED	1	RSS14 truncated
_SDK_RSS_BARCODE_TYPE_RSS14_STACKED	2	RSS14 stacked
_SDK_RSS_BARCODE_TYPE_RSS14_STACKED_OMNIDIRECTIONAL	3	RSS14 Stacked omnidirectional
_SDK_RSS_BARCODE_TYPE_RSS_LIMITED	4	RSS limited
_SDK_RSS_BARCODE_TYPE_RSS_EXPANDED	5	RSS Expanded
_SDK_RSS_BARCODE_TYPE_UPC_A	6	RSS UPC A
_SDK_RSS_BARCODE_TYPE_UPC_E	7	RSS UPC E
_SDK_RSS_BARCODE_TYPE_EAN13	8	EAN13
_SDK_RSS_BARCODE_TYPE_EAN8	9	EAN 8
_SDK_RSS_BARCODE_TYPE_UCC_EAN128_CC_A_B	10	EAN128 CC-A/B
_SDK_RSS_BARCODE_TYPE_UCC_EAN128_CC_C	11	EAN128 CC-C

magnification: magnification. *(range: 1~10)*

separatorHeight: Height of separator (*range:* 1~2)

barcodeHeight: this parameter only affects the UCC/EAN barcode type and CC-A/B/C barcode type.

segmentWidth: This parameter only affects the RSS expanded barcode type (*rang:* 0~22 Even number only)

rotation: Rotation

Code	Value	Description
_SDK_ROTATION_DEGREES_0	0	No rotation
_SDK_ROTATION_DEGREES_90	1	90 degrees of rotation
_SDK_ROTATION_DEGREES_180	2	180 degrees of rotation
_SDK_ROTATION_DEGREES_270	3	270 degrees of rotation.

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data



Contents requested by this API will be printed when **2-2-13 doPrint API** is called.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result =
[sdk drawBarcodeRSS:@"12345678901|this is composite info"
              xPosition:100
              yPosition:100
          barcodeType:_SDK_RSS_BARCODE_TYPE_RSS14
       magnification:2
    separatorHeight:1
       barcodeHeight:20
        segmentWidth:10
             rotation:_SDK_ROTATION_DEGREES_0];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-32 Draw Block

Draw line, Block, Box & Slope on the image buffer.

[Declaration]

```
-(__SDK_RESULT_CODES_) drawBlock:(NSInteger)startPosX
            startPosY:(NSInteger)startPosY
            endPosX:(NSInteger)endPosX
            endPosY:(NSInteger)endPosY
                option:(char)option
            thickness:(NSInteger)thickness
```

[Parameters]

startPosX: Horizontal start position (X)
(Reference: 1-3 Available range of X/Y coordinates for each model)
startPosY: Vertical start position (Y)
(Reference: 1-3 Available range of X/Y coordinates for each model)
endPosX: Horizontal end position (X)
(Reference: 1-3 Available range of X/Y coordinates for each model)
endPosY: Vertical end position (Y)
(Reference: 1-3 Available range of X/Y coordinates for each model)
option: Options

Code	Value	Description
_SDK_DRAW_BLOCK_OPTION_LINE_OVERWRITING	'O'	Line Overwriting
_SDK_DRAW_BLOCK_OPTION_LINE_EXCLUSIVE_OR	'E'	Line Exclusive OR
_SDK_DRAW_BLOCK_OPTION_LINE_DELETE	'D'	Line Delete
_SDK_DRAW_BLOCK_OPTION_SLOPE	'S'	Slope (a oblique line)
_SDK_DRAW_BLOCK_OPTION_BOX	'B'	Box

thickness: block thickness

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

Contents requested by this API will be printed when **2-2-13 doPrint API** is called.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result =
[sdk drawBarcodeRSS:@"12345678901|this is composite info"
              xPosition:100
              yPosition:100
          barcodeType:_SDK_RSS_BARCODE_TYPE_RSS14
        magnification:2
    separatorHeight:1
        barcodeHeight:20
         segmentWidth:10
               rotation:_SDK_ROTATION_DEGREES_0];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-33 Draw Circle

Draw Circle on the image buffer

[Declaration]

-(__SDK_RESULT_CODES_) drawCircle:(NSInteger)startPosX
startPosY:(NSInteger)startPosY
sizeSelection:(NSInteger)sizeSelection
multiplier:(NSInteger)multiplier

[Parameters]

startPosX: Horizontal start position (X)
(Reference: 1-3 Available range of X/Y coordinates for each model)
startPosY: Vertical start position (Y)
(Reference: 1-3 Available range of X/Y coordinates for each model)
sizeSelection: Circle Size Selection

Code	Value	Diameter (mm)	Width x Hegith(Dots)
_SDK_DRAW_CIRCLE_SIZE_40X40	1	5	40 × 40
_SDK_DRAW_CIRCLE_SIZE_56X56	2	7	56 × 56
_SDK_DRAW_CIRCLE_SIZE_72X72	3	9	72 × 72
_SDK_DRAW_CIRCLE_SIZE_88X88	4	11	88 × 88
_SDK_DRAW_CIRCLE_SIZE_104X104	5	13	104 × 104
_SDK_DRAW_CIRCLE_SIZE_168X168	6	21	168 × 168

multiplier: Multiplier(1~4)

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

Contents requested by this API will be printed when **2-2-13 doPrint API** is called.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result = [sdk drawCircle:100
                                      startPosY:100
                                sizeSelection:_SDK_DRAW_CIRCLE_SIZE_40X40
                                     multiplier:1];

if (result == _SDK_RESULT_SUCCESS) {
    ...
} else {
    ...
}
```

2-2-34 Draw Image

Draw image data on specific position of image buffer.
If you don't set image dithering, the default setting is 'NO'

[Declaration]

```
-(__SDK_RESULT_CODES_) drawImage:(UIImage*)image
                        startPosX:(NSInteger)startPosX
                        startPosY:(NSInteger)startPosY
                            width:(NSInteger)width
                            level:(NSUInteger)level
                    useDithering:(BOOL)useDithering
                    compressType:(__SDK_COMPRESS_TYPE)compressType
```

[Parameters]

Image : printing image
startPosX: Horizontal start position (X)
(Reference: 1-3 Available range of X/Y coordinates for each model)
startPosY: Vertical start position (Y)
(Reference: 1-3 Available range of X/Y coordinates for each model)
width: printing image width
(Reference: 1-3 Available range of X/Y coordinates for each model)
level: printing image brightness (0~100)
useDithering: Select to use image dithering
compressType: printing image compression type

Code	Value	Description
_SDK_COMPRESS_TYPE_NONE	0	Default
_SDK_COMPRESS_TYPE_RLE	1	RLE
_SDK_COMPRESS_TYPE_LZMA	2	LZMA

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_NO_CONNECT	0XF004	Printer is not connected
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data



Contents requested by this API will be printed when **2-2-13 doPrint API** is called.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

UIImage* img = [UIImage imageNamed:@"test"];

__SDK_RESULT_CODES_ result = [sdk drawImage:img
                                    startPosX:100
                                    startPosY:100
                                        width:300
level:50
useDithering:NO
compressType:_SDK_COMPRESS_TYPE_RLE
];

if (result == _SDK_RESULT_SUCCESS) {
    ...
} else {
    ...
}
```

2-2-35 Check Printer Status

Check printer status and report status data to host.

[Declaration]
-(long) checkPrinterStatus:(blockType1) statusReceiver;

[Parameters]
statusReceiver: It implements the function to be performed when the status of the printer is received.

Code	Value	Description
_SDK_STATUS_PRINTER_PAPEREMPTY	0x00000080	Paper empty
_SDK_STATUS_PRINTER_COVEROPEN	0x00000040	Cover open
_SDK_STATUS_PRINTER_CUTTER_JAMMED	0x00000020	Cutter jammed
_SDK_STATUS_PRINTER_TPH_OVERHEAT	0x00000010	Thermal head(TPH) overheat
_SDK_STATUS_GAP_ERROR	0x00000008	Gap detection error(Auto-sensing failure)
_SDK_STATUS_RIBBON_END	0x00000004	Ribbon end
_SDK_STATUS_PRINTER_BUILDING_LABEL	0x00008000	On building label to be printed in image buffer
_SDK_STATUS_PRINTER_PRINTING_LABEL	0x00004000	On printing label in image buffer
_SDK_STATUS_PRINTER_ISSUED_LABEL	0x00002000	Issued label is paused in peeler unit

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result = [sdk checkPrinterStatus:^(NSInteger parameter) {
    switch (parameter) {
        case _SDK_STATUS_PRINTER_COVEROPEN:
            NSLog(@"COVER OPEN");
            break;
        case _SDK_STATUS_PRINTER_GAP_ERROR:
            NSLog(@"GAP ERROR");
            break;
        case _SDK_STATUS_PRINTER_PAPEREMPTY:
            NSLog(@"Printer Paper Empty");
            break;
        case _SDK_STATUS_PRINTER_RIBBON_END:
            NSLog(@"Printer Ribbon End");
            break;
        case _SDK_STATUS_PRINTER_CUTTER_JAMMED :
            NSLog(@"Printer Cutter Jammed");
            break;
        case _SDK_STATUS_PRINTER_TPH_OVERHEAT :
            NSLog(@"Printer TPH Overheat");
            break;
        default:
            break;
    }
}];

if (result == _SDK_RESULT_SUCCESS) {
    ...
} else {
    ...
}
```

2-2-36 Set Auto Cutter: cutting Period

Enable or Disable auto-cut action after printing by doPrint method.

[Declaration]

-(__SDK_RESULT_CODES_) setAutoCutter:(BOOL)enableAutoCutter
cuttingPeriod:(NSInteger)cuttingPeriod

[Parameters]

enableAutoCutter: Cutting action enable/disable (YES: used NO: not used)
cuttingPeriod: Cutting period
Cutting period means the number of pages between two cuttings.

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data



This function is available only in models with Auto Cutter.
For models without Auto Cutter, always set the 'enableAutoCutter' to NO.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

// AutoCuter enable model
__SDK_RESULT_CODES_ result = [sdk setAutoCutter:YES cuttingPeriod:3];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-37 Get SDK Version String

It confirms the SDK version.

[Declaration]
-(NSString*) getSDKVersionString

[Return Value]
It returns SDK Version.

[Example]

```
NSLog(@"%@", [LabelPrinterSDK getSDKVersionString]);
```

2-2-38 Get Released Date String

It confirms the date when SDK was distributed.

[Declaration]
-(NSString*) getReleasedDateString

[Return Value]
It returns SDK release date.

[Example]

```
NSLog(@"%@", [LabelPrinterSDK getReleasedDateString]);
```

2-2-39 Get Device Font List

This method gets information and string values declared in relation to DeviceFont in NSMutableArray data type.

[Declaration]
-(NSMutableArray*) getDeviceFontList;

[Return Value]

Code	Value	Description
_SDK_DEVICE_FONT_6PT	'0'	9 X 15 (dots)
_SDK_DEVICE_FONT_8PT	'1'	12 X 20 (dots)
_SDK_DEVICE_FONT_10PT	'2'	16 X 25 (dots)
_SDK_DEVICE_FONT_12PT	'3'	19 X 30 (dots)
_SDK_DEVICE_FONT_15PT	'4'	24 X 38 (dots)
_SDK_DEVICE_FONT_20PT	'5'	32 X 40 (dots)
_SDK_DEVICE_FONT_30PT	'6'	48 X 76 (dots)
_SDK_DEVICE_FONT_14PT	'7'	22 X 34 (dots)
_SDK_DEVICE_FONT_18PT	'8'	28 X 44 (dots)
_SDK_DEVICE_FONT_24PT	'9'	37 X 58 (dots)
_SDK_DEVICE_FONT_KOREAN1	'a'	16 X 16 (dots) (ASCII 9 X 15)
_SDK_DEVICE_FONT_KOREAN2	'b'	24 X 24 (dots) (ASCII 12 X 24)
_SDK_DEVICE_FONT_KOREAN3	'c'	20 X 20 (dots) (ASCII 12 X 20)
_SDK_DEVICE_FONT_KOREAN4	'd'	26 X 26 (dots) (ASCII 16 X 30)
_SDK_DEVICE_FONT_KOREAN5	'e'	20 X 26 (dots) (ASCII 16 X 30)
_SDK_DEVICE_FONT_KOREAN6	'f'	38 X 38 (dots) (ASCII 22 X 34)
_SDK_DEVICE_FONT_GB2312	'm'	24 X 24 (dots) (ASCII 12 X 24)
_SDK_DEVICE_FONT_BIG5	'n'	24 X 24 (dots) (ASCII 12 X 24)
_SDK_DEVICE_FONT_SHIFT_JIS	'j'	24 X 24 (dots) (ASCII 12 X 24)

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

if (sdk.isConnected == NO) {
   return;
}

NSMutableArray* arrFontList = [sdk getDeviceFontList];
```

2-2-40 Get Vector Font List

This method gets information and string values declared in relation to VectorFont in NSMutableArray data type.

[Declaration]
-(NSMutableArray*) getVectorFontList

[Return Value]

Code	Value	Description
_SDK_VECTOR_FONT_ASCII	'U'	ASCII (1Byte code)
_SDK_VECTOR_FONT_KS5601	'K'	KS5601 (2Byte code)
_SDK_VECTOR_FONT_BIG5	'B'	BIG5 (2Byte code)
_SDK_VECTOR_FONT_GB2312	'G'	GB2312 (2Byte code)
_SDK_VECTOR_FONT_SHIFT_JIS	'J'	Shift-JIS (2Byte code)
_SDK_VECTOR_FONT_OCR_A	'a'	OCR-A (1Byte code)
_SDK_VECTOR_FONT_OCR_B	'b'	OCR-B (1Byte code)

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

if (sdk.isConnected == NO) {
    return;
}

NSMutableArray* arrFontList = [sdk getVectorFontList];
```

2-2-41 Get Draw Block Options

This method gets information and string values declared in relation to Draw Block Options in NSMutableArray data type.

[Declaration]
-(NSMutableArray*) getDrawBlockOptions;

[Return Value]

Code	Value	Description
_SDK_DRAW_BLOCK_OPTION_LINE_OVERWRITING	'O'	Line Overwriting
_SDK_DRAW_BLOCK_OPTION_LINE_EXCLUSIVE_OR	'E'	Line Exclusive OR
_SDK_DRAW_BLOCK_OPTION_LINE_DELETE	'D'	Line Delete
_SDK_DRAW_BLOCK_OPTION_SLOPE	'S'	Slope (a oblique line)
_SDK_DRAW_BLOCK_OPTION_BOX	'B'	Box

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

if (sdk.isConnected == NO) {
   return;
}

NSMutableArray* arrBlockOptionsList = [sdk getDrawBlockOptions];
```

2-2-42 Get Draw Circle Sizes

This method gets information and string values declared in relation to Draw Circle Sizes in NSMutableArray data type.

[Declaration]
-(NSMutableArray*) getDrawCircleSizes;

[Return Value]

Code	Value	Diameter (mm)	Width x Hegith(Dots)
_SDK_DRAW_CIRCLE_SIZE_40X40	1	5	40 × 40
_SDK_DRAW_CIRCLE_SIZE_56X56	2	7	56 × 56
_SDK_DRAW_CIRCLE_SIZE_72X72	3	9	72 × 72
_SDK_DRAW_CIRCLE_SIZE_88X88	4	11	88 × 88
_SDK_DRAW_CIRCLE_SIZE_104X104	5	13	104 × 104
_SDK_DRAW_CIRCLE_SIZE_168X168	6	21	168 × 168

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

if (sdk.isConnected == NO) {
    return;
}

NSMutableArray* arrCircleSizesList = [sdk getDrawCircleSizes];
```

2-2-43 Get Barcode Type 1D

This method gets information and string values declared in relation to 1D Barcode Type in NSMutableArray data type.

[Declaration]
-(NSMutableArray*) getBarcodeType1D;

[Return Value]

Code	Value	Description
_SDK_BARCODE_TYPE_CODE39	0	Code39
_SDK_BARCODE_TYPE_CODE128	1	Code128
_SDK_BARCODE_TYPE_I2Of5	2	Interleaved 2of5
_SDK_BARCODE_TYPE_CODABAR	3	Codabar
_SDK_BARCODE_TYPE_CODE93	4	Code93
_SDK_BARCODE_TYPE_UPC_A	5	UPC-A
_SDK_BARCODE_TYPE_UPC_E	6	UPC-E
_SDK_BARCODE_TYPE_EAN13	7	EAN13
_SDK_BARCODE_TYPE_EAN8	8	EAN8
_SDK_BARCODE_TYPE_EAN128	9	UCC/EAN128
_SDK_BARCODE_TYPE_CODE11	10	Code11
_SDK_BARCODE_TYPE_PLANET	11	Planet
_SDK_BARCODE_TYPE_INDUSTRIAL_2Of5	12	Industrial 2of5
_SDK_BARCODE_TYPE_STANDARD_2Of5	13	Standard 2of5
_SDK_BARCODE_TYPE_LOGMARS	14	Logmars
_SDK_BARCODE_TYPE_UPC_EAN_EXTENSIONS	15	UPC/EAN Extensions
_SDK_BARCODE_TYPE_POSTNET	16	Postnet

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

if (sdk.isConnected == NO) {
    return;
}

NSMutableArray* arrBarcodeType1D = [sdk getBarcodeType1D];
```

2-2-44 Get Barcode HRI

This method gets information and string values declared in relation to Barcode HRI in NSMutableArray data type.

[Declaration]
-(NSMutableArray*) getBarcodeHRI;

[Return Value]

Code	Value	Description
_SDK_BARCODE_HRI_NONE	0	HRI is not used.
_SDK_BARCODE_HRI_BELOW_FONTSIZE1	1	Position of HRI: Below barcode Font Size: 1
_SDK_BARCODE_HRI_ABOVE_FONTSIZE1	2	Position of HRI: Above barcode Font Size: 1
_SDK_BARCODE_HRI_BELOW_FONTSIZE2	3	Position of HRI: Below barcode Font Size: 2
_SDK_BARCODE_HRI_ABOVE_FONTSIZE2	4	Position of HRI: Above barcode Font Size: 2
_SDK_BARCODE_HRI_BELOW_FONTSIZE3	5	Position of HRI: Below barcode Font Size: 3
_SDK_BARCODE_HRI_ABOVE_FONTSIZE3	6	Position of HRI: Above barcode Font Size: 3
_SDK_BARCODE_HRI_BELOW_FONTSIZE4	7	Position of HRI: Below barcode Font Size: 4
_SDK_BARCODE_HRI_ABOVE_FONTSIZE4	8	Position of HRI: Above barcode Font Size: 4

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

if (sdk.isConnected == NO) {
      return;
}

NSMutableArray* arrBarcodeHRI = [sdk getBarcodeHRI];
```

2-2-45 Get Dpi

Check the DPI(Dot per Inch) value of the connected printer.

[Declaration]
-(NSInteger) getDpi;

[Return Value]
Return the DPI value set on the printer.
If not connected to the printer, return 0.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

NSInteger dpi = [sdk getDpi];
NSLog(@"DPI : %ld", dpi);
```

2-2-46 Get Max Width

Check the Maximum printable width on the connected printer.

[Declaration]
-(NSInteger) getMaxWidth;

[Return Value]
Return the maximum printable width value.
If not connected to the printer, return 0.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

NSInteger maxWidth = [sdk getMaxWidth];
NSLog(@"MAX WIDTH : %ld", maxWidth);
```

2-2-47 Set Orientation

Set printing direction.

[Declaration]
-(__SDK_RESULT_CODES_) setOrientation:(NSInteger)orientation

[Parameters]
orientation: Printing direction

Code	Value	Description
_SDK_ORIENTATION_TOP_BOTTOM	0	Print from top to bottom
_SDK_ORIENTATION_BOTTOM_TOP	1	Print from bottom to top

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result =
                              [sdk setOrientation:_SDK_ORIENTATION_TOP_BOTTOM];

if (result == _SDK_RESULT_SUCCESS) {
    ...
} else {
    ...
}
```

2-2-48 Set ICS: code page

To select international character set and code table

[Declaration]
-(__SDK_RESULT_CODES_) setICS:(NSInteger)ics codepage:(NSInteger)cp

[Parameters]
ics: International character set

Code	Value	Description
_SDK_CONFIG_ICS_USA	0	U.S.A
_SDK_CONFIG_ICS_FRANCE	1	France
_SDK_CONFIG_ICS_GERMANY	2	Germany
_SDK_CONFIG_ICS_UK	3	U.K
_SDK_CONFIG_ICS_DENMARK_I	4	Denmark I
_SDK_CONFIG_ICS_SWEDEN	5	Sweden
_SDK_CONFIG_ICS_ITALY	6	Italy
_SDK_CONFIG_ICS_SPAIN_I	7	Spain I
_SDK_CONFIG_ICS_NORWAY	8	Norway
_SDK_CONFIG_ICS_DENMARK_II	9	Denmark II
_SDK_CONFIG_ICS_JAPAN	10	Japan
_SDK_CONFIG_ICS_SPAIN_II	11	Spain II
_SDK_CONFIG_ICS_LATIN_AMERICA	12	Latin America
_SDK_CONFIG_ICS_KOREA	13	Korea
_SDK_CONFIG_ICS_SLOVENIA_CROATIA	14	Slovenia/Croatia
_SDK_CONFIG_ICS_CHINA	15	China

cp: codepage

Code	Value	Description	
_SDK_CONFIG_CODEPAGE_CP437	0	CP437	U.S.A
_SDK_CONFIG_CODEPAGE_CP850	1	CP850	Latin1
_SDK_CONFIG_CODEPAGE_CP852	2	CP 852	Latin2
_SDK_CONFIG_CODEPAGE_CP860	3	CP 860	Portuguese
_SDK_CONFIG_CODEPAGE_CP863	4	CP 863	Canadian French
_SDK_CONFIG_CODEPAGE_CP865	5	CP 865	Nordic
_SDK_CONFIG_CODEPAGE_CP1252	6	WCP 1252	Latin I
_SDK_CONFIG_CODEPAGE_CP865_WCP1252	7	CP 865 + WCP 1252	European Combined
_SDK_CONFIG_CODEPAGE_CP857	8	CP 857	Turkish
_SDK_CONFIG_CODEPAGE_CP737	9	CP 737	Greek
_SDK_CONFIG_CODEPAGE_WPC1250	10	WCP 1250	Latin 2
_SDK_CONFIG_CODEPAGE_WPC1253	11	WCP 1253	Greek
_SDK_CONFIG_CODEPAGE_WPC1254	12	WCP 1254	Turkish
_SDK_CONFIG_CODEPAGE_CP855	13	CP 855	Cyrillic
_SDK_CONFIG_CODEPAGE_CP862	14	CP 862	Hebrew
_SDK_CONFIG_CODEPAGE_CP866	15	CP 866	Cyrillic
_SDK_CONFIG_CODEPAGE_WCP1251	16	WCP 1251	Cyrillic
_SDK_CONFIG_CODEPAGE_WCP1255	17	WCP 1255	Hebrew
_SDK_CONFIG_CODEPAGE_CP928	18	CP 928	Greek
_SDK_CONFIG_CODEPAGE_CP864	19	CP 864	Arabic
_SDK_CONFIG_CODEPAGE_CP775	20	CP 775	Baltic
_SDK_CONFIG_CODEPAGE_WCP1257	21	WCP1257	Baltic
_SDK_CONFIG_CODEPAGE_CP858	22	CP858	Latin 1 + Euro

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result = [sdk setICS:_SDK_CONFIG_ICS_USA
                                codepage:_SDK_CONFIG_CODEPAGE_CP437];

if (result == _SDK_RESULT_SUCCESS) {
    ...
} else {
    ...
}
```

2-2-49 Set Printing Type

Select Thermal Direct Printing or Thermal Transfer Printing.

[Declaration]
-(__SDK_RESULT_CODES_) setPrintingType:(char)printingType

[Parameters]
printingType: Direct Thermal / Thermal transfer

Code	Value	Description
_SDK_PRINTING_TYPE_DIRECT_THERMAL	'd'	Direct thermal
_SDK_PRINTING_TYPE_THERMAL_TRANSFER	't'	Thermal transfer

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result =
    [sdk setPrintingType:_SDK_PRINTING_TYPE_DIRECT_THERMAL];

if (result == _SDK_RESULT_SUCCESS) {
    ...
} else {
    ...
}
```

2-2-50 Set Margin: vertical Margin

Set marginal value of the image buffer.
This method moves the origin point and make become the new origin.

[Declaration]

```
-(__SDK_RESULT_CODES_) setMargin:(NSInteger)horizontalMargin
                        verticalMargin:(NSInteger)verticalMargin
```

[Parameters]

horizontalMargin: Horizontal margin
verticalMargin: Vertical margin

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result = [sdk setMargin:10
                                verticalMargin:10];

if (result == _SDK_RESULT_SUCCESS) {
    ...
} else {
    ...
}
```

2-2-51 Set Length

Set length of label and gap(or Black mark) and specify media type.

[Declaration]

```
-(__SDK_RESULT_CODES_) setLength:(NSInteger) labelLength
          gepLength:(NSInteger) gapLength
          mediaType:(char) mediaType
        offsetLength:(NSInteger)offsetLength
```

[Parameters]

labelLength: Label length
gapLength: Gap length or thickness of black line
mediaType: Media type

Code	Value	Description
_SDK_MEDIA_TYPE_GAP	'G'	Gap
_SDK_MEDIA_TYPE_CONTINUOUS	'C'	Continuous
_SDK_MEDIA_TYPE_BLACK_MARK	'B'	Black mark

offsetLength: Offset Length between Black mark(or Gap) and perforation line

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];
...

__SDK_RESULT_CODES_ result = [sdk setLength:1216
                                  gepLength:1
                                  mediaType:_SDK_MEDIA_TYPE_BLACK_MARK
                               offsetLength:1];

if (result == _SDK_RESULT_SUCCESS) {
   ...
} else {
   ...
}
```

2-2-52 Set Rewinder

It activates Rewinder for models with Rewinder.

[Declaration]
-(__SDK_RESULT_CODES_) setRewinder:(BOOL)enableRewinder
[Parameters]
enableRewinder: It chooses whether or not to use Rewinder.

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data



This function is available only in models with Rewinder.
For models without Rewinder, always set the 'enableRewinder' to NO.

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

// Rewinder enable model
__SDK_RESULT_CODES_ result = [sdk setRewinder:YES];

if (result == _SDK_RESULT_SUCCESS) {
      ...
} else {
      ...
}
```

2-2-53 Set Speed

Set print speed.

[Declaration]
-(__SDK_RESULT_CODES_) setSpeed:(NSInteger) speedValue

[Parameters]
speedValue: speed set value

Code	Value	Description
_SDK_SPEED_VALUE_1_0_IPS	7	1.0 Inch print per second
_SDK_SPEED_VALUE_2_0_IPS	8	2.0 Inch print per second
_SDK_SPEED_VALUE_2_5_IPS	0	2.5 Inch print per second
_SDK_SPEED_VALUE_3_0_IPS	1	3.0 Inch print per second
_SDK_SPEED_VALUE_4_0_IPS	2	4.0 Inch print per second
_SDK_SPEED_VALUE_5_0_IPS	3	5.0 Inch print per second
_SDK_SPEED_VALUE_6_0_IPS	4	6.0 Inch print per second
_SDK_SPEED_VALUE_7_0_IPS	5	7.0 Inch print per second
_SDK_SPEED_VALUE_8_0_IPS	6	8.0 Inch print per second
_SDK_SPEED_VALUE_9_0_IPS	9	9.0 Inch print per second
_SDK_SPEED_VALUE_10_0_IPS	10	10.0 Inch print per second
_SDK_SPEED_VALUE_11_0_IPS	11	11.0 Inch print per second
_SDK_SPEED_VALUE_12_0_IPS	12	12.0 Inch print per second
_SDK_SPEED_VALUE_13_0_IPS	13	13.0 Inch print per second
_SDK_SPEED_VALUE_14_0_IPS	14	14.0 Inch print per second

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result = [sdk setSpeed:_SDK_SPEED_VALUE_8_0_IPS];

if (result == _SDK_RESULT_SUCCESS) {
    ...
} else {
    ...
}
```

2-2-54 Set Offset

Save (set) offset length between black marks(or gap) and dotted lines

[Declaration]
-(__SDK_RESULT_CODES_) setOffset:(NSInteger)offsetValue

[Parameters]
offsetValue: offset (range: -100~100)

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result = [sdk setOffset:1];

if (result == _SDK_RESULT_SUCCESS) {
    ...
} else {
    ...
}
```

2-2-55 Set Density

Set print Density.

[Declaration]
-(__SDK_RESULT_CODES_) setDensity:(NSInteger) densityValue

[Parameters]
densityValue: Density level (range: 0~20)

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result = [sdk setDensity:10];

if (result == _SDK_RESULT_SUCCESS) {
    ...
} else {
    ...
}
```

2-2-56 Set Cutting Position

This method regulates the label cutting location after printing.
Tear-off position or Cut position can adjust.

[Declaration]
-(__SDK_RESULT_CODES_) setCuttingPosition:(NSInteger)cuttiongPosition

[Parameters]
cuttiongPosition: tear-off/cutter position(range: -100~100)

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result = [sdk setCuttingPosition:1];

if (result == _SDK_RESULT_SUCCESS) {
    ...
} else {
    ...
}
```

2-2-57 Set Width

Set label width.
Resize the image buffer to match the label size.

[Declaration]
-(__SDK_RESULT_CODES_) setWidth:(NSInteger) widthValue

[Parameters]
widthValue: label width
(Reference: 1-3 Available range of X/Y coordinates for each model)

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result = [sdk setWidth:811];

if (result == _SDK_RESULT_SUCCESS) {
    ...
} else {
    ...
}
```

2-2-58 Print Raw Data

Executes the printing by commands to the printer.
Please refer to SLCS(Programming) Manual for command generation.

[Declaration]
-(__SDK_RESULT_CODES_) printRawData:(NSData*)rawData;

[Parameters]
rawData: Instruction data generated by SLCS

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

unsigned char command[] = { 0x53, 0x53, 0x33, 0x0d, 0x10, 0x00 };
NSData *data = [NSData dataWithBytes:&command length:sizeof(command)];

__SDK_RESULT_CODES_ result = [sdk printRawData:data];

if (result == _SDK_RESULT_SUCCESS) {
      NSLog(@"SUCCESS");
} else {
      NSLog(@"FAIL");
}
```

2-2-59 Get Model Name

Check the Model Name of the connected printer.

[Declaration]
-(NSString *) getModelName;

[Return Value]
Printer model name

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

NSString* modelName = [sdk getModelName];

NSLog(modelName);
```

2-2-60 Get Firmware Version

Check the Firmware Version of the connected printer.

[Declaration]
-(NSString *) getFirmwareVersion;

[Return Value]
Printer firmware version

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

NSString* fwVer = [sdk getFirmwareVersion];

NSLog(fwVer);
```

2-2-61 Setup RFID

Set the RFID transponder type, coding position, number of coding retries, number of labels upon retry, and sending/receiving power.

[Declaration]

```
-(__SDK_RESULT_CODES_) setupRFIDWithRfidType:(NSInteger) rfidType
                                numberOfRetries:(NSInteger) numberOfRetries
                                  numberOfLabel:(NSInteger) numberOfLabel
                                     radioPower:(NSInteger) radioPower
```

[Parameters]

rfidType: RFID Transponder Type (value area is 0~5 (
0 = none,
1 = ISO_18000_6_Type_A,
2 = ISO_18000_6_Type_B,
3 = Class_0,
4 = Class_1,
5 = Class_1_Gen 2 (default)
))
numberOfRetries: Number of Coding Retries Upon Coding Failure (1~10)
numberOfLabel: Number of Labels Upon Retry Following RFID Label Writing Failure (1~5)
radioPower: Sending/Receiving Power Adjustment (0~30)

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result = [printer setupRFIDWithRfidType:0
                                                numberOfRetries:2
                                                  numberOfLabel:3
                                                     radioPower:5];
NSLog(result);
```

2-2-62 Set RFID Position

For calculating and saving the optimal coding position of the RFID label on the printer and printing

[Declaration]
-(__SDK_RESULT_CODES_) setRFIDPositionWithPosition:(NSInteger) position

[Parameters]
position: Coding Position (Y-Axis Value)

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result = [sdk setRFIDPositionWithPosition:30];

NSLog(result);
```

2-2-63 Write RFID

For reading or writing RFID labels

[Declaration]

```
-(__SDK_RESULT_CODES_) writeRFIDWithDataType:(NSInteger) dataType
                                  startingBlockNumber:(NSInteger) startingBlockNumber
                                           dataLength:(NSInteger) dataLength
                                                 data:(NSString*) data
```

[Parameters]

dataType: Data Type(value area is 0~3 (
0 = ASCII,
1 = Hex,
2 = EPC,
3 = User Hex
))
startingBlockNumber: Starting Block Number (4~10)
dataLength: Number of Bytes for Reading (2~12)
data: Write Data

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result = [sdk writeRFIDWithDataType:0
                                          startingBlockNumber:4
                                                   dataLength:5
                                                         data:@”12345”];
NSLog(result);
```

2-2-64 Set RFID Password

For setting the RFID access password and kill password

[Declaration]

```
-(__SDK_RESULT_CODES_)setRFIDPasswordWithOldAccessPwd
    :(NSString*)oldAccessPwd
    oldKillPwd:(NSString*) oldKillPwd
    newAccessPwd:(NSString*) newAccessPwd
    newKillPwd:(NSString*) newKillPwd
```

[Parameters]

oldAccessPwd: Old Access Password 4byte (Currently Active Access Password)
oldKillPwd: Old Kill Password 4byte (Currently Active Kill Password)
newAccessPwd: New Access Password 4byte (Modified Access Password)
newKillPwd: New Kill Password 4byte (Modified Kill Password)

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result = [sdk
      setRFIDPasswordWithOldAccessPwd:@”00004444”
      oldKillPwd: @”00004444”
      newAccessPwd: @”EFEB1299”
      newKillPwd: @”EFEB1299”];
NSLog(result);
```

2-2-65 Lock RFID

For locking data

[Declaration]
-(__SDK_RESULT_CODES_) lockRFID;

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

...

__SDK_RESULT_CODES_ result = [sdk lockRFID];

NSLog(result);
```

2-2-66 Set EPC Data Structure

sets up the RFID EPC data structure

[Declaration]

-(__SDK_RESULT_CODES_) setEPCDataStructureWithTotalSize:(NSInteger)
totalSize
fieldSizes:(NSString*) fieldSizes

[Parameters]

totalSize: total size
fieldSizes: size per field

[Return Value]

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data

[Example]

```
LabelPrinterSDK* sdk = [LabelPrinterSDK new];

…

__SDK_RESULT_CODES_ result = [sdk setEPCDataStructureWithTotalSize:@”10”

fieldSizes:@”1,2.3.4”]

NSLog(result);
```

2-2-67 Set Keep Network Connection

Sets whether to keep the connection to the network (Ethernet, Wi-Fi) printer or not.

[Declaration]

-(void) setKeepNetworkConnection:(NSInteger)enable;

[Parameters]

enable: 0 or negative to disconnect, positive to keep connected



If there is no data communication between the network (Ethernet, Wi-Fi) printer and the HOST for the specified time (Inactivity Time), the connection is automatically terminated. Call this method to keep the connection alive regardless of this time. This method must be called before the 'connectWithAddress', 'connectWithTarget', and 'connectWithSerialNumber' methods.

3. Constants

Constants used in the provided SDK are defined in the “LabelPrinterSDK_Defines.h” file.

3-1 Result Codes

• Results Codes are the values returned from various methods, which indicate error conditions of the corresponding method.

Code	Value	Description
_SDK_RESULT_SUCCESS	0X0000	Success
_SDK_RESULT_FAIL	0XF000	Fail
_SDK_RESULT_FAIL_NOT_SUPPORT_FUNCTION	0XF001	Not supported
_SDK_RESULT_FAIL_NO_OPEN	0XF002	SDK is not open
_SDK_RESULT_FAIL_OPEN_ALREADY	0XF003	SDK is already open
_SDK_RESULT_FAIL_NO_CONNECT	0XF004	Printer is not connected
_SDK_RESULT_FAIL_CONNECT_ALREADY	0XF005	Printer is already connected
_SDK_RESULT_FAIL_WRITE_ERROR	0XF006	Failed to send data
_SDK_RESULT_FAIL_READ_ERROR	0XF007	Failed to receive data
_SDK_RESULT_FAIL_INVALID_PARAMETER	0XF008	Invalid parameter

3-2 Alignment

• Alignment constants are used to specify the alignment property.

Code	Value	Description
_SDK_ALIGNMENT_LEFT	0	Align to left
_SDK_ALIGNMENT_RIGHT	1	Align to right
_SDK_ALIGNMENT_CENTER	2	Align to center
_SDK_ALIGNMENT_STRING_FROM_RIGHT_2_LEFT	2	Print characters from right to left

_SDK_ALIGNMENT_CENTER: Only valid for “Vector Fonts”.
_SDK_ALIGNMENT_STRING_FROM_RIGHT_2_LEFT: Only valid for Device Font.

3-3 Barcode HRI

• The below constants are used to specify the position and font of HRI when printing barcodes that supports HRI.

Code	Value	Description
_SDK_BARCODE_HRI_NONE	0	HRI is not used.
_SDK_BARCODE_HRI_BELOW_FONTSIZE1	1	Position of HRI: Below barcode Font Size: 1
_SDK_BARCODE_HRI_ABOVE_FONTSIZE1	2	Position of HRI: Above barcode Font Size: 1
_SDK_BARCODE_HRI_BELOW_FONTSIZE2	3	Position of HRI: Below barcode Font Size: 2
_SDK_BARCODE_HRI_ABOVE_FONTSIZE2	4	Position of HRI: Above barcode Font Size: 2
_SDK_BARCODE_HRI_BELOW_FONTSIZE3	5	Position of HRI: Below barcode Font Size: 3
_SDK_BARCODE_HRI_ABOVE_FONTSIZE3	6	Position of HRI: Above barcode Font Size: 3
_SDK_BARCODE_HRI_BELOW_FONTSIZE4	7	Position of HRI: Below barcode Font Size: 4
_SDK_BARCODE_HRI_ABOVE_FONTSIZE4	8	Position of HRI: Above barcode Font Size: 4

3-4 Maxi Code Modes

• Maxi Code Mode constants are used to set the barcode option when printing Maxi code barcode.

Code	Value	Description
_SDK_MAXICODE_MODE_0	0	MaxiCode Mode 0
_SDK_MAXICODE_MODE_2	2	MaxiCode Mode 2
_SDK_MAXICODE_MODE_3	3	MaxiCode Mode 3
_SDK_MAXICODE_MODE_4	4	MaxiCode Mode 4

3-5 1D Barcode Types

• One-dimensional Barcode Types constants are used to set the barcode option when printing 1D barcode.

Code	Value	Description
_SDK_BARCODE_TYPE_CODE39	0	Code39
_SDK_BARCODE_TYPE_CODE128	1	Code128
_SDK_BARCODE_TYPE_I2Of5	2	Interleaved 2of5
_SDK_BARCODE_TYPE_CODABAR	3	Codabar
_SDK_BARCODE_TYPE_CODE93	4	Code93
_SDK_BARCODE_TYPE_UPC_A	5	UPC-A
_SDK_BARCODE_TYPE_UPC_E	6	UPC-E
_SDK_BARCODE_TYPE_EAN13	7	EAN13
_SDK_BARCODE_TYPE_EAN8	8	EAN8
_SDK_BARCODE_TYPE_EAN128	9	UCC/EAN128
_SDK_BARCODE_TYPE_CODE11	10	Code11
_SDK_BARCODE_TYPE_PLANET	11	Planet
_SDK_BARCODE_TYPE_INDUSTRIAL_2Of5	12	Industrial 2of5
_SDK_BARCODE_TYPE_STANDARD_2Of5	13	Standard 2of5
_SDK_BARCODE_TYPE_LOGMARS	14	Logmars
_SDK_BARCODE_TYPE_UPC_EAN_EXTENSIONS	15	UPC/EAN Extensions
_SDK_BARCODE_TYPE_POSTNET	16	Postnet

3-6 Barcode Origin Point

• Barcode Origin Point constants are used to set the reference origin position of barcode.

Code	Value	Description
_SDK_BARCODE_ORIGIN_POINT_CENTER	0	Set the reference point of the barcode to the center.
_SDK_BARCODE_ORIGIN_POINT_UpperLeft	1	Set the reference point of the barcode in the upper left corner.

3-7 Error Correction Level

• Error Correction Level constants are used to set the level of error correction for possible corruption of barcode.

Code	Value	Description
_SDK_ERROR_CORRECTION_LEVEL0	0	Error Correction Level 0
_SDK_ERROR_CORRECTION_LEVEL1	1	Error Correction Level 1
_SDK_ERROR_CORRECTION_LEVEL2	2	Error Correction Level 2
_SDK_ERROR_CORRECTION_LEVEL3	3	Error Correction Level 3
_SDK_ERROR_CORRECTION_LEVEL4	4	Error Correction Level 4
_SDK_ERROR_CORRECTION_LEVEL5	5	Error Correction Level 5
_SDK_ERROR_CORRECTION_LEVEL6	6	Error Correction Level 6
_SDK_ERROR_CORRECTION_LEVEL7	7	Error Correction Level 7
_SDK_ERROR_CORRECTION_LEVEL8	8	Error Correction Level 8

3-8 Data Compression Method

• Data Compression Method constants are used to specify the data compression property.

Code	Value	Description
_SDK_DATA_COMPRESSION_METHOD_TEXT	0	2char / codeword
_SDK_DATA_COMPRESSION_METHOD_NUMERIC	1	2.93 char / codeword
_SDK_DATA_COMPRESSION_METHOD_BINARY	2	1.2bytes / codeword

3-9 QR Code Model

• QR Code Model constants are used to set the options in printing QR barcode.

Code	Value	Description
_SDK_QRCODE_MODEL_1	1	QR Model 1
_SDK_QRCODE_MODEL_2	2	QR Model 2

3-10 Code 49 Starting Mode

• Code 49 Starting Mode constants are used to set the properties of Starting Mode in printing Code 49 barcode.

Code	Value	Description
_SDK_STARTINGMODE_REGULAR_ALPHANUMERIC	0	Regular Alphanumeric Mode
_SDK_STARTINGMODE_MULTIPLE_READ_ALPHANUMERIC	1	Multiple Read Alphanumeric
_SDK_STARTINGMODE_REGULAR_NUMERIC	2	Regular Numeric Mode
_SDK_STARTINGMODE_GROUP_ALPHANUMERIC	3	Group Alphanumeric Mode
_SDK_STARTINGMODE_REGULAR_ALPHANUMRIC_SHIFT1	4	Regular Alphanumeric Shift 1
_SDK_STARTINGMODE_REGULAR_ALPHANUMRIC_SHIFT2	5	Regular Alphanumeric Shift 2
_SDK_STARTINGMODE_AUTOMATIC	7	Automatic Mode

3-11 Coda block Mode

• Codablock Mode constants are used to set the options when printing Codablock barcode.

Code	Value	Description
_SDK_CODABLACK_MODE_A	'A'	Code 39 character set is used.
_SDK_CODABLACK_MODE_E	'E'	Code 128 character set is used.
_SDK_CODABLACK_MODE_F	'F'	Code 128 character set is used. Function 1 is added automatically.

3-12 Check Digit Option

• These constants are used to set the Check Digit property when printing MSI barcode.

Code	Value	Description
_SDK_CHECKDIGIT_NONE	0	No Check Digit
_SDK_CHECKDIGIT_1MOD10	1	Check Digit 1 Mod 10
_SDK_CHECKDIGIT_2MOD10	2	Check Digit 2 Mod 10
_SDK_CHECKDIGIT_1MOD11_AND_1MOD_10	3	Check Digit 1 Mod 10

3-13 RSS Barcode Type

• RSS Barcode Type constants are used to set the barcode type when printing RSS barcode.

Code	Value	Description
_SDK_RSS_BARCODE_TYPE_RSS14	0	RSS14
_SDK_RSS_BARCODE_TYPE_RSS14_TRUNCATED	1	RSS14 truncated
_SDK_RSS_BARCODE_TYPE_RSS14_STACKED	2	RSS14 stacked
_SDK_RSS_BARCODE_TYPE_RSS14_STACKED_OMNIDIRECTIONAL	3	RSS14 Stacked omnidirectional
_SDK_RSS_BARCODE_TYPE_RSS_LIMITED	4	RSS limited
_SDK_RSS_BARCODE_TYPE_RSS_EXPANDED	5	RSS Expanded
_SDK_RSS_BARCODE_TYPE_UPC_A	6	RSS UPC A
_SDK_RSS_BARCODE_TYPE_UPC_E	7	RSS UPC E
_SDK_RSS_BARCODE_TYPE_EAN13	8	EAN13
_SDK_RSS_BARCODE_TYPE_EAN8	9	EAN 8
_SDK_RSS_BARCODE_TYPE_UCC_EAN128_CC_A_B	10	EAN128 CC-A/B
_SDK_RSS_BARCODE_TYPE_UCC_EAN128_CC_C	11	EAN128 CC-C

3-14 Rotation Degrees

• Rotation Degrees constants are used to set the rotation property of the printing.

Code	Value	Description
_SDK_ROTATION_DEGREES_0	0	No rotation
_SDK_ROTATION_DEGREES_90	1	90 degrees of rotation
_SDK_ROTATION_DEGREES_180	2	180 degrees of rotation
_SDK_ROTATION_DEGREES_270	3	270 degrees of rotation.

3-15 Device Fonts

• Device Fonts constants are used to set the property of Device Font.

Code	Value	Description
_SDK_DEVICE_FONT_6PT	'0'	9 X 15 (dots)
_SDK_DEVICE_FONT_8PT	'1'	12 X 20 (dots)
_SDK_DEVICE_FONT_10PT	'2'	16 X 25 (dots)
_SDK_DEVICE_FONT_12PT	'3'	19 X 30 (dots)
_SDK_DEVICE_FONT_15PT	'4'	24 X 38 (dots)
_SDK_DEVICE_FONT_20PT	'5'	32 X 40 (dots)
_SDK_DEVICE_FONT_30PT	'6'	48 X 76 (dots)
_SDK_DEVICE_FONT_14PT	'7'	22 X 34 (dots)
_SDK_DEVICE_FONT_18PT	'8'	28 X 44 (dots)
_SDK_DEVICE_FONT_24PT	'9'	37 X 58 (dots)
_SDK_DEVICE_FONT_KOREAN1	'a'	16 X 16 (dots) (ASCII 9 X 15)
_SDK_DEVICE_FONT_KOREAN2	'b'	24 X 24 (dots) (ASCII 12 X 24)
_SDK_DEVICE_FONT_KOREAN3	'c'	20 X 20 (dots) (ASCII 12 X 20)
_SDK_DEVICE_FONT_KOREAN4	'd'	26 X 26 (dots) (ASCII 16 X 30)
_SDK_DEVICE_FONT_KOREAN5	'e'	20 X 26 (dots) (ASCII 16 X 30)
_SDK_DEVICE_FONT_KOREAN6	'f'	38 X 38 (dots) (ASCII 22 X 34)
_SDK_DEVICE_FONT_GB2312	'm'	24 X 24 (dots) (ASCII 12 X 24)
_SDK_DEVICE_FONT_BIG5	'n'	24 X 24 (dots) (ASCII 12 X 24)
_SDK_DEVICE_FONT_SHIFT_JIS	'j'	24 X 24 (dots) (ASCII 12 X 24)

3-16 Vector Fonts

• Vector Fonts constants are used to set the property of Vector Fonts.

Code	Value	Description
_SDK_VECTOR_FONT_ASCII	'U'	ASCII (1Byte code)
_SDK_VECTOR_FONT_KS5601	'K'	KS5601 (2Byte code)
_SDK_VECTOR_FONT_BIG5	'B'	BIG5 (2Byte code)
_SDK_VECTOR_FONT_GB2312	'G'	GB2312 (2Byte code)
_SDK_VECTOR_FONT_SHIFT_JIS	'J'	Shift-JIS (2Byte code)
_SDK_VECTOR_FONT_OCR_A	'a'	OCR-A (1Byte code)
_SDK_VECTOR_FONT_OCR_B	'b'	OCR-B (1Byte code)

3-17 Draw Block Options

• Draw Block Options constants are used to set the Draw Block Options.

Code	Value	Description
_SDK_DRAW_BLOCK_OPTION_LINE_OVERWRITING	'O'	Line Overwriting
_SDK_DRAW_BLOCK_OPTION_LINE_EXCLUSIVE_OR	'E'	Line Exclusive OR
_SDK_DRAW_BLOCK_OPTION_LINE_DELETE	'D'	Line Delete
_SDK_DRAW_BLOCK_OPTION_SLOPE	'S'	Slope(a oblique line)
_SDK_DRAW_BLOCK_OPTION_BOX	'B'	Box

3-18 Draw Circle Sizes

• Draw Circle Sizes constants are used to set the property related to the size when using the Draw Circle Method.

Code	Value	Diameter (mm)	Width x Hegith(Dots)
_SDK_DRAW_CIRCLE_SIZE_40X40	1	5	40 × 40
_SDK_DRAW_CIRCLE_SIZE_56X56	2	7	56 × 56
_SDK_DRAW_CIRCLE_SIZE_72X72	3	9	72 × 72
_SDK_DRAW_CIRCLE_SIZE_88X88	4	11	88 × 88
_SDK_DRAW_CIRCLE_SIZE_104X104	5	13	104 × 104
_SDK_DRAW_CIRCLE_SIZE_168X168	6	21	168 × 168

3-19 International Character Set

• These constants are used to set the International Character Set.

Code	Value	Description
_SDK_CONFIG_ICS_USA	0	U.S.A
_SDK_CONFIG_ICS_FRANCE	1	France
_SDK_CONFIG_ICS_GERMANY	2	Germany
_SDK_CONFIG_ICS_UK	3	U.K
_SDK_CONFIG_ICS_DENMARK_I	4	Denmark I
_SDK_CONFIG_ICS_SWEDEN	5	Sweden
_SDK_CONFIG_ICS_ITALY	6	Italy
_SDK_CONFIG_ICS_SPAIN_I	7	Spain I
_SDK_CONFIG_ICS_NORWAY	8	Norway
_SDK_CONFIG_ICS_DENMARK_II	9	Denmark II
_SDK_CONFIG_ICS_JAPAN	10	Japan
_SDK_CONFIG_ICS_SPAIN_II	11	Spain II
_SDK_CONFIG_ICS_LATIN_AMERICA	12	Latin America
_SDK_CONFIG_ICS_KOREA	13	Korea
_SDK_CONFIG_ICS_SLOVENIA_CROATIA	14	Slovenia/Croatia
_SDK_CONFIG_ICS_CHINA	15	China

3-20 Code Pages

• These constants are used to set the Code Page.

Code	Value	Description	
_SDK_CONFIG_CODEPAGE_CP437	0	CP437	U.S.A
_SDK_CONFIG_CODEPAGE_CP850	1	CP850	Latin1
_SDK_CONFIG_CODEPAGE_CP852	2	CP 852	Latin2
_SDK_CONFIG_CODEPAGE_CP860	3	CP 860	Portuguese
_SDK_CONFIG_CODEPAGE_CP863	4	CP 863	Canadian French
_SDK_CONFIG_CODEPAGE_CP865	5	CP 865	Nordic
_SDK_CONFIG_CODEPAGE_CP1252	6	WCP 1252	Latin I
_SDK_CONFIG_CODEPAGE_CP865_WCP1252	7	CP 865 + WCP 1252	European Combined
_SDK_CONFIG_CODEPAGE_CP857	8	CP 857	Turkish
_SDK_CONFIG_CODEPAGE_CP737	9	CP 737	Greek
_SDK_CONFIG_CODEPAGE_WPC1250	10	WCP 1250	Latin 2
_SDK_CONFIG_CODEPAGE_WPC1253	11	WCP 1253	Greek
_SDK_CONFIG_CODEPAGE_WPC1254	12	WCP 1254	Turkish
_SDK_CONFIG_CODEPAGE_CP855	13	CP 855	Cyrillic
_SDK_CONFIG_CODEPAGE_CP862	14	CP 862	Hebrew
_SDK_CONFIG_CODEPAGE_CP866	15	CP 866	Cyrillic
_SDK_CONFIG_CODEPAGE_WCP1251	16	WCP 1251	Cyrillic
_SDK_CONFIG_CODEPAGE_WCP1255	17	WCP 1255	Hebrew
_SDK_CONFIG_CODEPAGE_CP928	18	CP 928	Greek
_SDK_CONFIG_CODEPAGE_CP864	19	CP 864	Arabic
_SDK_CONFIG_CODEPAGE_CP775	20	CP 775	Baltic
_SDK_CONFIG_CODEPAGE_WCP1257	21	WCP1257	Baltic
_SDK_CONFIG_CODEPAGE_CP858	22	CP858	Latin 1 + Euro

3-21 Printing Type

• These constants are used to set the Printing type.

Code	Value	Description
_SDK_PRINTING_TYPE_DIRECT_THERMAL	'd'	Direct thermal
_SDK_PRINTING_TYPE_THERMAL_TRANSFER	't'	Thermal transfer

3-22 Media Type

• These constants are used to set the print media type.

Code	Value	Description
_SDK_MEDIA_TYPE_GAP	'G'	Gap
_SDK_MEDIA_TYPE_CONTINUOUS	'C'	Continuous
_SDK_MEDIA_TYPE_BLACK_MARK	'B'	Black mark

3-23 Speed Value

• These constants are used to set the print speed.

Code	Value	Description
_SDK_SPEED_VALUE_1_0_IPS	7	1.0 Inch print per second
_SDK_SPEED_VALUE_2_0_IPS	8	2.0 Inch print per second
_SDK_SPEED_VALUE_2_5_IPS	0	2.5 Inch print per second
_SDK_SPEED_VALUE_3_0_IPS	1	3.0 Inch print per second
_SDK_SPEED_VALUE_4_0_IPS	2	4.0 Inch print per second
_SDK_SPEED_VALUE_5_0_IPS	3	5.0 Inch print per second
_SDK_SPEED_VALUE_6_0_IPS	4	6.0 Inch print per second
_SDK_SPEED_VALUE_7_0_IPS	5	7.0 Inch print per second
_SDK_SPEED_VALUE_8_0_IPS	6	8.0 Inch print per second
_SDK_SPEED_VALUE_9_0_IPS	9	9.0 Inch print per second
_SDK_SPEED_VALUE_10_0_IPS	10	10.0 Inch print per second
_SDK_SPEED_VALUE_11_0_IPS	11	11.0 Inch print per second
_SDK_SPEED_VALUE_12_0_IPS	12	12.0 Inch print per second
_SDK_SPEED_VALUE_13_0_IPS	13	13.0 Inch print per second
_SDK_SPEED_VALUE_14_0_IPS	14	14.0 Inch print per second

3-24 Orientation

• These constants are used to set the printing direction.

Code	Value	Description
_SDK_ORIENTATION_TOP_BOTTOM	0	Print from top to bottom
_SDK_ORIENTATION_BOTTOM_TOP	1	Print from bottom to top

3-25 Encoding Options

• These constants are used to set the Text encoding.

Code	Value	Description
_SDK_ENCODING_OPTION_AUTOMATIC	0xFFFFFFFF	Auto
_SDK_ENCODING_OPTION_ASCII	0x00000001	ASCII
_SDK_ENCODING_OPTION_KSC5601	0x80000422	Korean graphic character set for information interchange
_SDK_ENCODING_OPTION_SHIFT_JIS	0x00000008	Shift Japanese industrial standards
_SDK_ENCODING_OPTION_BIG5	0x80000a03	Traditional Chinese characters
_SDK_ENCODING_OPTION_GB2312	0x80000421	Simplified Chinese characters(94x94)
_SDK_ENCODING_OPTION_GB18030	0x80000632	Simplified Chinese characters

3-26 Printer Status

• These constants are used to check the printer error status.

Code	Value	Description
_SDK_STATUS_PRINTER_PAPEREMPTY	0x00000080	Paper empty
_SDK_STATUS_PRINTER_COVEROPEN	0x00000040	Cover open
_SDK_STATUS_PRINTER_CUTTER_JAMMED	0x00000020	Cutter jammed
_SDK_STATUS_PRINTER_TPH_OVERHEAT	0x00000010	Thermal head(TPH) overheat
_SDK_STATUS_GAP_ERROR	0x00000008	Gap detection error (Auto-sensing failure)
_SDK_STATUS_RIBBON_END	0x00000004	Ribbon end
_SDK_STATUS_PRINTER_BUILDING_LABEL	0x00008000	On building label to be printed in image buffer
_SDK_STATUS_PRINTER_PRINTING_LABEL	0x00004000	On printing label in image buffer
_SDK_STATUS_PRINTER_ISSUED_LABEL	0x00002000	Issued label is paused in peeler unit

3-27 Compress Type

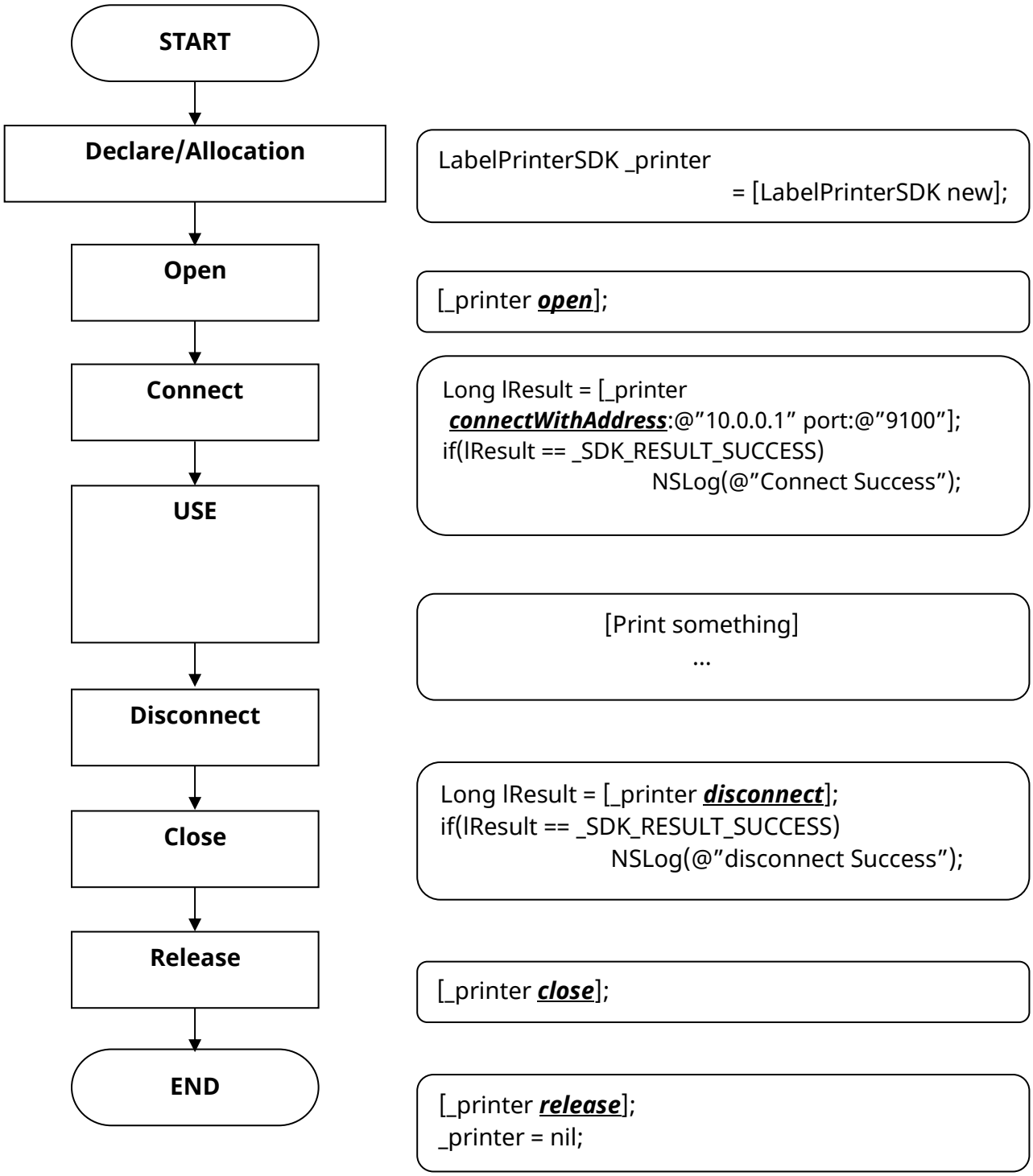
• These constants are used to set the image compression type.

Code	Value	Description
_SDK_COMPRESS_TYPE_NONE	0	Default
_SDK_COMPRESS_TYPE_RLE	1	RLE
_SDK_COMPRESS_TYPE_LZMA	2	LZMA

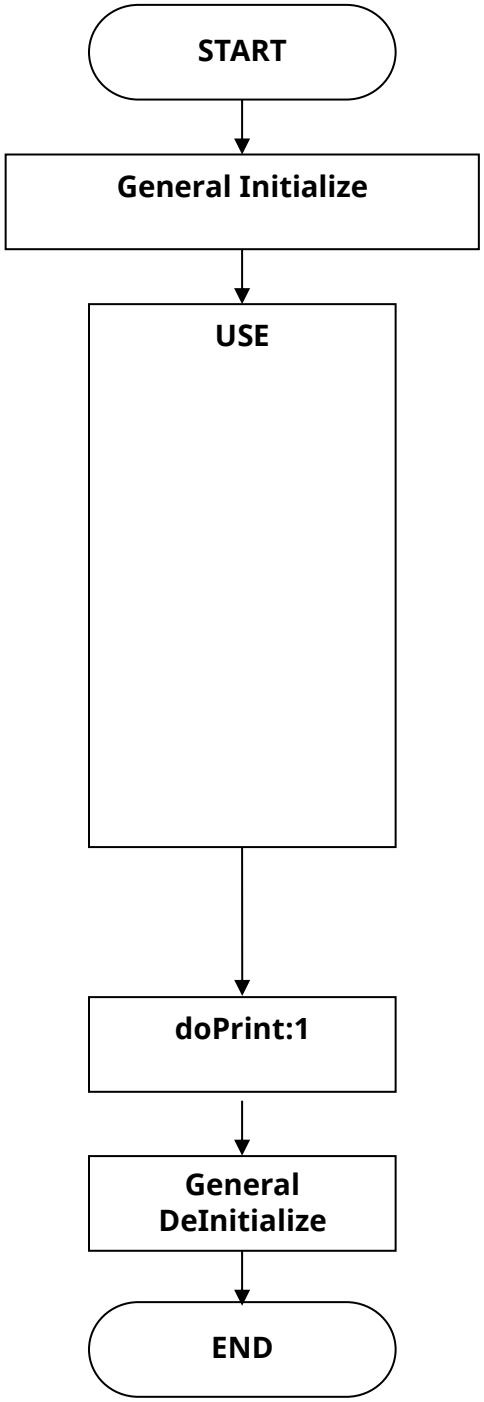
4. Appendix

4-1 Recommended Flow of Operation

4-1-1 General



4-1-2 Printing Contents



General ***Initialize*** ... (Declare ~ Connect)
Reference: 4-1-1 General

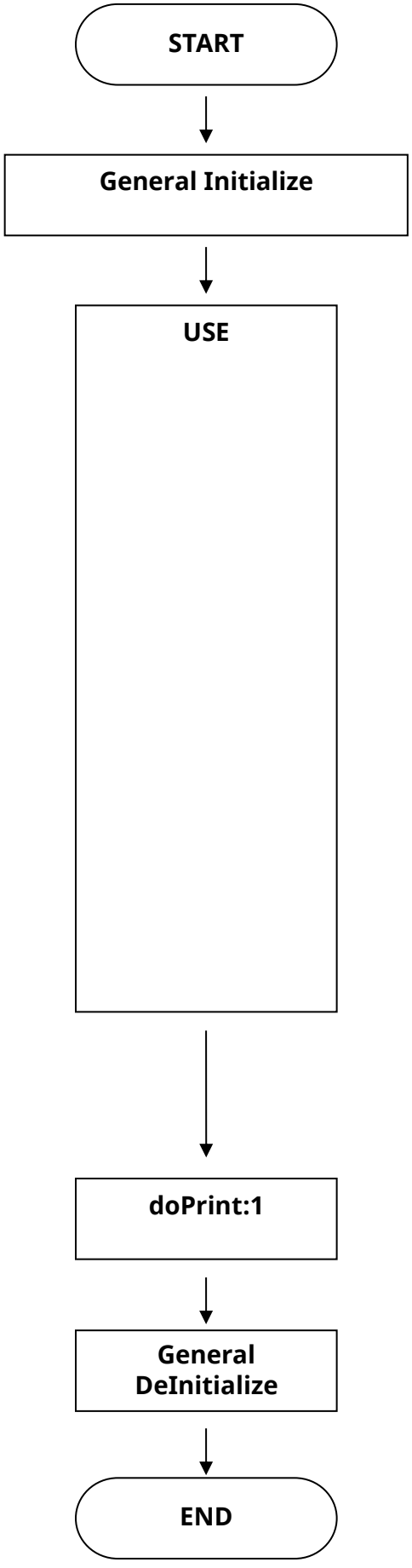
```
// [USE] Functions to print barcode, text, image,
and various shapes can be used.
Long lResult =
[_printer drawImage: _imageSample.image
            startPosX:100
            startPosY:100
                 width:300];

if(lResult = _SDK_RESULT_SUCCESS)
                    NSLog(@"drawImage Success");
```

```
Long lResult =    [_printer doPrint:1];
if(lResult = _SDK_RESULT_SUCCESS)
                    NSLog(@"drawImage Success");
```

General ***DeInitialize*** ... (Declare ~ Connect)
Reference: 4-1-1 General

4-1-3 Printing Two or More Contents on One Side



General ***Initialize*** ... (Declare ~ Connect)
Reference: 4-1-1 General

```
//    [Contents 1]
Long lResult =
[_printer drawImage: _imageSample.image
            startPosX:100
            startPosY:100
                 width:300];
if(lResult = _SDK_RESULT_SUCCESS)
                  NSLog(@"drawImage Success");
//    [Contents 2]
lResult =
[_printer  drawTextDeviceFont:@"0*  12pt,  70,
220" xPosition:70 yPosition:220
fontSelection:_SDK_DEVICE_FONT_12PT
fontWidth:1 fontHeight:1
rightSideCharacterSpacing:0
fontRotation:_SDK_ROTATION_DEGREES_0
reverse:NO bold:NO
textAlignment:_SDK_ALIGNMENT_LEFT];
if(lResult = _SDK_RESULT_SUCCESS)
                     NSLog(@"drawText Success");
```

```
Long lResult =    [_printer doPrint:1];
if(lResult = _SDK_RESULT_SUCCESS)
                   NSLog(@"drawImage Success");
```

General ***DeInitialize*** ... (Declare ~ Connect)
Reference: 4-1-1 General

4-2 Sample Codes

4-2-1 Connecting/Disconnecting

```
// Declaration/Allocation and open SDK
LabelPrinterSDK _printer =    [LabelPrinterSDK new];
[_printer open];
//     Can be selected between <1> and <2>.
/*Start Of <1>*/
// Connect to the printer that is connected with IP Address “10.0.0.1” and Port 9100
long lResult = [_printer    connectWithAddress:@”10.0.0.1” port:@”9100”];
if(lResult == _SDK_RESULT_SUCCESS)
NSLog(@”Connect Success”); /*Start Of <1>*/

/*Start Of <2>*/
//    Attempt to connect to printer assigned to IP Address “10.0.0.1” Port 9100
long lResult = 0;

      LabelPrinterObject* prt = [LabelPrinterObject new];

        [prt setAddress:@”10.0.0.1”];
        [prt setPort:@”9100”];
        [prt setInterfaceType:_SDK_INTERFACETYPE_WIFI];
        lResult = [_printer connectWithTarget:prt timeout:3000];
      [_parent          alert:[NSString          stringWithFormat:@"onConnect          %s",
g_resultString[(lResult == _SDK_RESULT_SUCCESS)]]];/*Start Of <2>*/


// After finishing the use of SDK
lResult = [_printer disconnect]; // Disconnect
[_printer release];
_printer = nil;
```

4-2-2 Printing Contents

```
long   lResult = [_printer doPrint:1]; // Any number bigger than 1 will repeat the same printing for the specified number of times.

if(lResult == _SDK_RESULT_SUCCESS)
      NSLog(@"doPrint Success");
```

4-2-3 Printing a Rectangular Box

```
lResult = [_printer drawBlock:100
startPosY:100
endPosX:300 endPosY:300
option:_SDK_DRAW_BLOCK_OPTION_SLOPE
thickness:1];

NSLog(@" drawBlock Success");
```

4-2-4 Printing a Circle

```
lResult = [_printer drawCircle:200
startPosY:200
sizeSelection:_SDK_DRAW_CIRCLE_SIZE_40X40
multiplier:1];

if(lResult == _SDK_RESULT_SUCCESS)
NSLog(@"drawTextDeviceFont Success");
```

4-2-5 Printing a Text String with Device Font

```
lResult = [_printer drawTextDeviceFont:@"device Font Test"
xPosition:70    yPosition:220
fontSelection:_SDK_DEVICE_FONT_12PT
fontWidth:1 fontHeight:1
rightSideCharacterSpacing:0
fontRotation:_SDK_ROTATION_DEGREES_0
reverse:NObold:NO textAlignment:_SDK_ALIGNMENT_LEFT];
if(lResult == _SDK_RESULT_SUCCESS)
NSLog(@”drawTextDeviceFont Success”);
```

4-2-6 Printing a Text String with Vector Font

```
lResult = [_printer drawTextVectorFont:@"draw Font Test - Vector"
  xPosition:50 yPosition:100
fontSelection:_SDK_VECTOR_FONT_ASCII
fontWidth:10 fontHeight:10
rightSideCharacterSpacing:0
fontRotation:_SDK_ROTATION_DEGREES_0
reverse:NO bold:YES italic:YES
textWriteDirectionRightToLeft:NO textAlignment:_SDK_ALIGNMENT_LEFT];

if(lResult == _SDK_RESULT_SUCCESS)
NSLog(@”drawTextVectorFont Success”);
```

4-2-7 Printing 1D Barcode

```
lResult = [_printer drawBarcode1D:@">A1234567890"
xPosition:50 yPosition:100
barcodeType:_SDK_BARCODE_TYPE_CODE128
widthNarrow:2 widthWide:6 height:100
hri:_SDK_BARCODE_HRI_BELOW_FONTSIZE1
quietZoneWidth:0
rotation:_SDK_ROTATION_DEGREES_0];

if(lResult == _SDK_RESULT_SUCCESS)
      NSLog(@”drawBarcode1D Success”);
```

4-2-8 Printing PDF417

```
lResult =[_printer drawBarcodePDF417:@"PDF417 Test"
xPosition:50 yPosition:300
maximumRowCount:30 maximumColumnCount:5
errorCorrectionLevel:_SDK_ERROR_CORRECTION_LEVEL0
dataCompressionMethod:_SDK_DATA_COMPRESSION_METHOD_TEXT
printBarcodeText:YES
barcodeOriginPoint:_SDK_BARCODE_ORIGIN_POINT_UpperLeft
moduleWidth:3 barHeight:10
rotation:_SDK_ROTATION_DEGREES_0];

if(lResult == _SDK_RESULT_SUCCESS)
NSLog(@”drawBarcodePDF417 Success”);
```

4-2-9 Printing QR Cod

```
lResult =[_printer drawBarcodeQRCode:@"QR Code Test"
xPosition:450 yPosition:300
barcodeSize:3
model:_SDK_QRCODE_MODEl_2
errorColectionLevel:_SDK_ERROR_CORRECTION_LEVEL1
rotation:_SDK_ROTATION_DEGREES_0];

if(lResult == _SDK_RESULT_SUCCESS)
NSLog(@”drawBarcodeQRCode Success”);
```

4-2-10 Printing Data Matrix

```
lResult =[_printer drawBarcodeDataMatrix:@"dataMatrix Test"
xPosition:600 yPosition:300
barcodeSize:3 reverse:NO
rotation:_SDK_ROTATION_DEGREES_0];

if(lResult == _SDK_RESULT_SUCCESS)
NSLog(@” drawBarcodeDataMatrix Success”);
```

4-2-11 Printing Maxi Code

```
lResult = [_printer drawBarcodeMaxiCode:@"Maxi Code Test!!!"
xPosition:50 yPosition:400
mode:_SDK_MAXICODE_MODE_0];

if(lResult == _SDK_RESULT_SUCCESS)
NSLog(@”drawBarcodeMaxiCode Success”);
```

4-2-12 Printing Aztec

```
lResult = [_printer drawBarcodeAztec:@"Aztec Barcode Test"
xPosition:450 yPosition:400
barcodeSize:3
extendedChannel:NO
errorCorrectionLevel:0
menuSymbol:NO numberOfSymbols:1
optionalID:@""
rotation:_SDK_ROTATION_DEGREES_0];

if(lResult == _SDK_RESULT_SUCCESS)
NSLog(@”drawBarcodeAztec Success”);
```

4-2-13 Printing Code49

```
lResult = [_printer drawBarcodeCode49:@"Code 49 Test"
xPosition:600 yPosition:400
widthNarrow:1 widthWide:3
height:100
hri:_SDK_BARCODE_HRI_ABOVE_FONTSIZE1
startingMode:_SDK_STARTINGMODE_AUTOMATIC
rotation:_SDK_ROTATION_DEGREES_0];

if(lResult == _SDK_RESULT_SUCCESS)
NSLog(@” drawBarcodeCode49 Success”);
```

4-2-14 Printing Coda block

```
lResult = [_printer drawBarcodeIMB:@"01234567890123456789"
xPosition:50 yPosition:1000
printBarcodeText:YES
rotation:_SDK_ROTATION_DEGREES_0];

if(lResult == _SDK_RESULT_SUCCESS)
NSLog(@" drawBarcodeCodaBlock Success");
```

4-2-15 Printing Micro PDF417

```
lResult = [_printer drawBarcodeMicroPDF:@"MICRO PDF 417 TEST"
xPosition:50 yPosition:900
moduleWidth:4
barcodeHeight:4
mode:8
rotation:_SDK_ROTATION_DEGREES_0];

if(lResult == _SDK_RESULT_SUCCESS)
NSLog(@"drawBarcodeMicroPDF Success");
```

4-2-16 Printing IMB

```
lResult = [_printer drawBarcodeIMB:@"01234567890123456789"
xPosition:50 yPosition:1000
printBarcodeText:YES
rotation:_SDK_ROTATION_DEGREES_0];

if(lResult == _SDK_RESULT_SUCCESS)
NSLog(@" drawBarcodeIMB Success");
```

4-2-17 Printing MSI

```
lResult = [_printer drawBarcodeMSI:@"123456"
xPosition:50 yPosition:1100
widthNarrow:2 widthWide:7
height:100
checkDigitSelection:_SDK_CHECKDIGIT_1MOD10
printCheckDigitInHRI:YES
hri:_SDK_BARCODE_HRI_BELOW_FONTSIZE1
rotation:_SDK_ROTATION_DEGREES_0];

if(lResult == _SDK_RESULT_SUCCESS)
NSLog(@" drawBarcodeMSI Success");
```

4-2-18 Printing Plessey

```
lResult = [_printer drawBarcodePlessey:@"12345"
xPosition:100 yPosition:100
widthNarrow:2 widthWide:7
height:100 printCheckDigit:YES
hri:_SDK_BARCODE_HRI_BELOW_FONTSIZE1
rotation:_SDK_ROTATION_DEGREES_0];

if(lResult == _SDK_RESULT_SUCCESS)
NSLog(@" drawBarcodePlessey Success");
```

4-2-19 Printing TLC39

```
lResult = [_printer drawBarcodeTLC39:@"123456,ABCD12345678901234"
xPosition:100 yPosition:300
widthNarrow:2 widthWide:7
height:50 rowHeightOfMicroPDF417:3
narrowWidthOfMicroPDF417:2
rotation:_SDK_ROTATION_DEGREES_0];

if(lResult == _SDK_RESULT_SUCCESS)
NSLog(@" drawBarcodeTLC39 Success");
```

4-2-20 Printing RSS

```
lResult = [_printer drawBarcodeRSS:@"12345678901|this is composite info"
xPosition:100 yPosition:600
barcodeType:_SDK_RSS_BARCODE_TYPE_RSS14
magnification:2
separatorHeight:1 barcodeHeight:20 segmentWidth:10
rotation:_SDK_ROTATION_DEGREES_0];

if(lResult == _SDK_RESULT_SUCCESS)
NSLog(@" drawBarcodeRSS Success");
```

4-2-21 Bluetooth Printer Serch

```
NSArray* arrayPrinter = [_printer getPairedDevices];
    for(LabelPrinterObject* printer in arrayPrinter)
    {
        [_textViewSearchResult setText:[NSString
stringWithFormat:@"%@\r\n%@", _textViewSearchResult.text,
                                        [NSString stringWithFormat:@"[%@]
IF:%@/Serial:%@", [printer getModelName], @"BT(Clasic)",[printer
getSerialNumber]]]];

    }
```

4-2-22 Searching BLE / Wi-Fi / Ethernet Printer

```
//   Lookup
[_printer lookupPrinterWithCount:5 interval:0.1];

///   Delegate Process.
- (void)didFindPrinter:(LabelPrinterSDK *)controller
                printer:(LabelPrinterObject *)printer
{    //   Called every time a printer is detected.

     if([printer getInterfaceType] == _SDK_INTERFACETYPE_BLUETOOTH_BLE){
          NSLog(@"didFindPrinter(BLE) : %@", [printer getMacAddress]);
     }
     else{
          NSLog(@"didFindPrinter(WIFI/ETH) : %@", [printer getAddress]);
     }
}

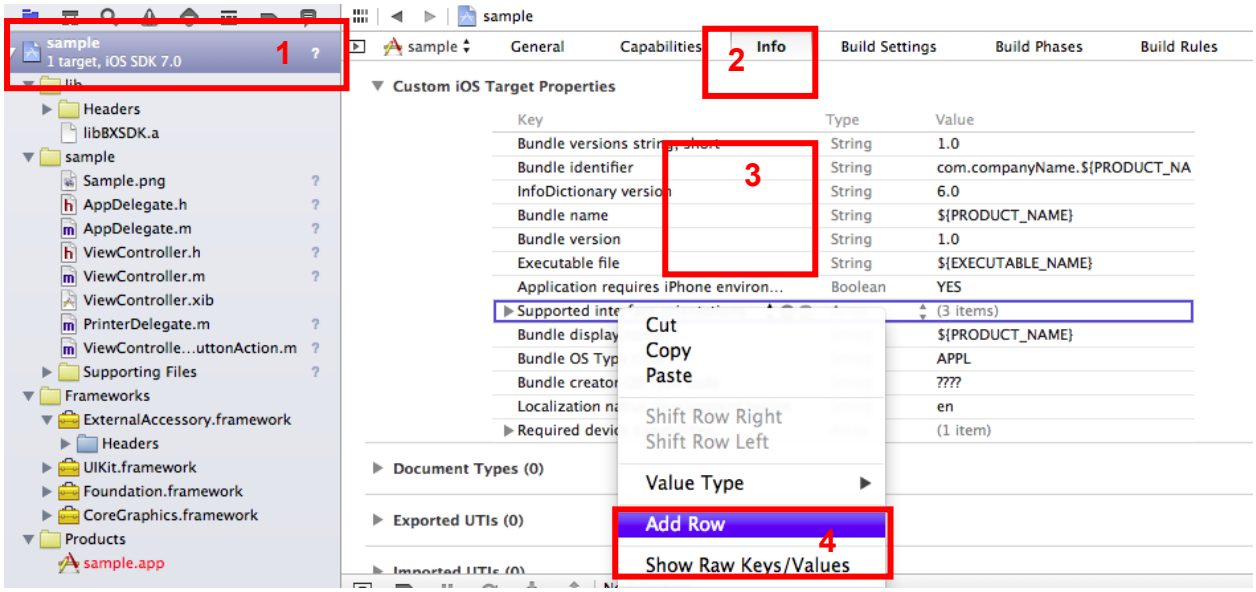
- (void)didLookupPrinters:(LabelPrinterSDK *)controller
                printerList:(NSArray *)printerList
{   //   Called after printer search is all completed.

     NSLog(@"didLookupPrinter");
//      [self alert:@"did Lookup Find Printer"];
}
```

4-3 Setting SDK Project

• Registration with the Apple Developer Program is required to develop iOS applications. Refer to the Apple Developer's Website (http://developer.apple.com/devcenter/ios) for detail

4-3-1 Adding Bluetooth Protocol

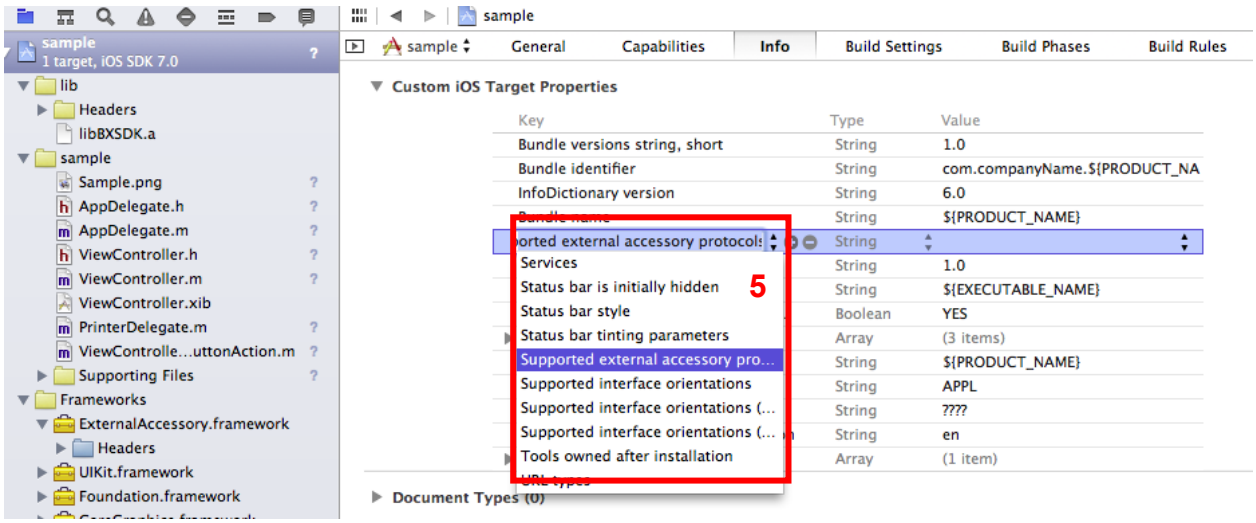


1) Select project file

2) Select [Info] tab

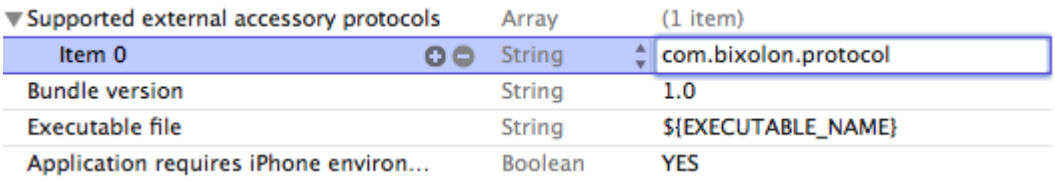
3) Ctrl-click in the Area 3

4) Select [Add Row] from the pop up menu.



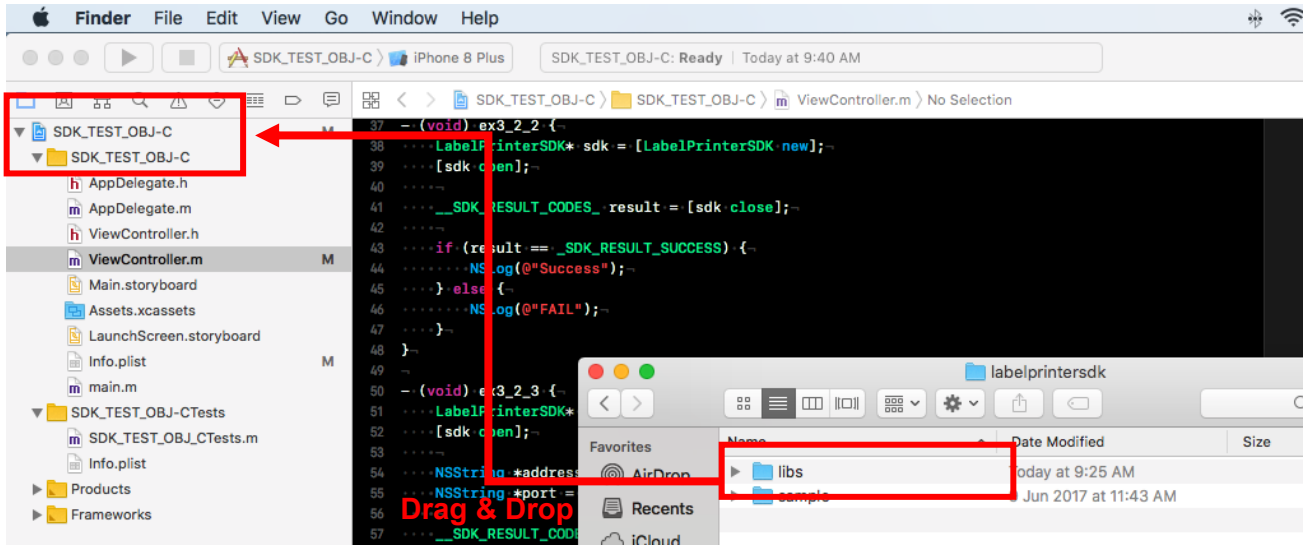
5) Enter “Supported external accessory protocols”

6) Enter “com.bixolon.protocol” in the items field.



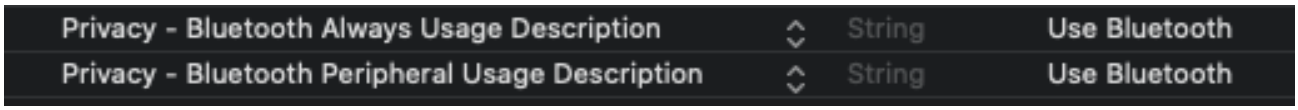
4-3-2 Adding Library to Projct

Drag the library folder and drop it to the project.



4-3-3 Add Bluetooth permission to Info.plist

- iOS 10, or later you need to add the 'Privacy – Bluetooth Peripheral Usage Description' item.
- iOS 13, or later you need to add the 'Privacy – Bluetooth Always Usage Description' item.



4-3-4 Multicast Networking Entitlement Request

If the iOS project deployment target version is 12 or higher or the iPhone/iPad OS version is 14 or higher, you must need to request multicast permission from Apple to discover printer devices. See the link below.

1) https://developer.apple.com/news/?id=0oi77447

2) https://developer.apple.com/contact/request/networking-multicast



- IP address used for discovering devices: 255.255.255.255
- Port numbers used for discovering devices: 48780, 48781, 9000, 3337

Revision history

Ver.	Date	Description
3.00	14-05-2018	New
3.01	21-01-2019	add supported model and method. add 'drawImage' parameter. add 'checkPrinterStatus' method response result value. add model name, firmware version check method modify project settings.
3.02	06-03-2019	add supported model modify QR code barcode size
3.03	25-06-2019	modify supported model
3.04	31-10-2019	modify supported models and speeds Remove sample program description
3.05	08-11-2019	add supported model Add Bluetooth permission settings
3.06	12-03-2020	add supported model(XM7-20)
3.07	07-04-2020	add supported image compress type
3.08	11-09-2020	add RFID type model (XT5-40(RFID)/XT5-43(RFID)/XT5-46(RFID), XD5-40t(RFID)/XD5-43t(RFID)) add supported RFID API
3.09	20-10-2020	add supported model(SRP-S3000)
3.10	12-03-2021	1) The table for supported printers updated. 2) Typo error fixed in 'drawImage' API
3.11	19-08-2021	1) Tables for specifications of the supported printers updated. 2) 'setKeepNetworkConnection' added 3) 'Multicast Networking Entitlement Request' added
3.12	20-04-2022	1) Supported iOS version changed. 2) Tables for specifications of the supported printers updated.
3.13	28-10-2024	1) Updates on printer spec sheet and new models - New Models - XD5-40IId - XD5-43IId - XD5-40IIt - XD5-43IIt
3.14	11-04-2025	1) Updates on printer spec sheet and new models New Models - XM5-30(RFID) - BT3-40/ BT-43 - BT5-40(RFID) / BT5-43(RFID) / BT5-46(RFID)

		- BD5-40d/ BD5-43d - BD5-40t(RFID)/ BD5-43t(RFID)
3.15	10-09-2025	1) Updates on printer spec sheet and new models New Models - XT6-60N / XT6-63N
3.16	12-08-2026	1) Updates on printer spec sheet and new models New Models - XC3-40d - XC5-40d - XC5-43d